On the Impossibility of Lossless Convexification for Attitude-Rate-Limited (5-DOF and 6-DOF) Powered Descent

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Abstract

Lossless convexification (LCvx) solves the three-degree-of-freedom (3-DOF) powered-descent guidance problem, including the nonconvex minimum-thrust constraint, by one convex program. Its relaxed solutions are provably feasible for the original problem. We ask whether the same is possible once the thrust direction is tied to a vehicle attitude that turns at a finite rate. This is the case for every thrust-vector-controlled (TVC) lander. We give a negative answer for the class of relaxations that LCvx belongs to: convex programs in which the physical state is a decision variable and the boundary data enter affinely. No such relaxation can be lossless for 5-DOF (translation plus thrust-axis pointing) or rate-limited 6-DOF descent. The obstruction is geometric. A lossless relaxation of this kind forces the set of feasible boundary conditions to be convex (Theorem A). For attitude-rate-limited models this set is not convex. The reason is that the vehicle cannot chatter its thrust direction. We prove this without state constraints, with altitude and glide-slope constraints for every horizon, for upright landings, for free final time on compact windows, for line search over the final time, with or without a minimum throttle, and for kinematic and independent-torque 6-DOF models. A per-instance version shows that every state-linear relaxation of a minimum-landing-error problem is lossy on explicit feasible instances, and an optimization version shows that no uniform relaxation is value-lossless for all convex costs. On the positive side, LCvx is exact once the attitude profile is fixed. For the gimbal-torque-coupled TVC rigid body, the problem with a given attitude profile is exactly convex, with no relaxation needed. So all of the nonconvexity lies in the choice of attitude profile. Degree-2 semidefinite lifts are not exact in general, but they give a posteriori global-optimality and infeasibility certificates, and they are certified exact on a coplanar subclass. The core obstruction, its state-constrained version, and the abstract per-instance and line-search results are machine-checked in Lean 4/Mathlib. Numerically, the obstruction is thin for in-air descents of a small lander (centimetre scale), but reaches metre scale for pad-launched hops.

Internal draft conventions (remove before submission)

Status tags after theorem titles: [Lean-checked] machine-checked (no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound` only); [proved; survived attack] rigorous proof that survived an adversarial review; [proved in project notes; registry status CLAIMED] a written proof exists and the orchestrator's synthesis lists it as proved, but the claims registry has not yet been moved to CONFIRMED; [numerical evidence only]; [open]. The mapping theorem → registry row → proof file → Lean name is in `paper/CLAIMS-MAP.md`. Toggle with `\showstatusfalse`.

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1 Introduction

Lossless convexification. Powered-descent guidance asks for a thrust history that takes a lander from its current position and velocity to rest at a target, subject to thrust-magnitude bounds $0 < \rho_1 \le \|T\| \le \rho_2$, pointing and state constraints, and often minimum fuel or minimum landing error. The lower thrust bound makes the control set an annulus, which is nonconvex. The lossless convexification (LCvx) programme of Açıkmeşe and coauthors [1–5] replaces the annulus by a convex cone in an augmented space, $\|T\| \le \Gamma$, $\rho_1 \le \Gamma \le \rho_2$, and proves that optimal solutions of the relaxed second-order cone program (SOCP) satisfy $\|T\| = \Gamma$ almost everywhere. They are therefore optimal for the original problem. Extensions cover pointing constraints [4, 6], state constraints [7, 8], some nonlinear dynamics [9], normality-based general theory [10], free final time [11] and discrete time [12]. Surveys and tutorials are [13, 14]. LCvx-based guidance has flown [15, 16]. Every one of these results treats the thrust direction as a free control variable, possibly restricted to a cone about a fixed or prescribed axis. Six-degree-of-freedom (6-DOF) powered-descent methods use sequential convex programming instead [17–21], and LCvx appears there only as an embedded relaxation of the thrust bounds [14]. A search of the literature through November 2026 found no LCvx theorem for a model in which the thrust direction is a rate-limited state, and no impossibility result for lossless convexification of any kind.

Why 5-DOF and 6-DOF matter. On a gimballed (TVC) lander the thrust acts along an engine axis that can deflect by at most a gimbal angle δ from the body axis b. The body axis itself turns at a finite rate, and on a pure-TVC vehicle it is turned *by* the gimbal. A 3-DOF LCvx solution can ask for thrust-direction changes the vehicle cannot follow. The lander that motivated this study is a small throttleable TVC vehicle ($\approx$ 80 kg wet, thrust 400–1000 N, $\pm 15^\circ$ box gimbal, usable body rate 0.3–0.5 rad/s). Its stored 3-DOF LCvx solutions contain thrust-direction jumps of 15° within one 24 ms control step, plus 180° azimuth flips, which would require angular accelerations around 10^3 rad/s^2. The vehicle can deliver 1–3.5 rad/s^2. The question we address is whether *any* LCvx-type convex program can be lossless once the attitude dynamics are part of the model.

The answer in one paragraph. 3-DOF LCvx works because the reachable set of a linear system with a measurable control is convex: whatever the relaxed problem does on average can be realized by fast switching (Lyapunov–Aumann [22, 23]). In 5-DOF the thrust direction is tied to a body axis that rotates at a finite rate, so it cannot switch. Take a maximal tilt to the left and a maximal tilt to the right. The average of these two manoeuvres is not flyable. Any convex program that contains both must also contain their average, provided the physical state is a decision variable and the boundary data enter affinely. That is exactly the LCvx situation. The rest of the paper makes this precise, removes the idealizations one by one (state constraints, minimum throttle, terminal attitude, final time, 6-DOF), quantifies the size of the obstruction, and describes what can still be convexified exactly.

Contributions.

1. *Definitions* (Section 2). We define uniform and state-linear convex relaxations with affine (or separable-monotone) data entry, and lossless relaxations in a feasibility (detection-sound) form and in an optimization form. We show that the standard LCvx definition with attainment implies the feasibility form. We also state precisely what the class excludes: occupation-measure and moment lifts with nonaffine data entry, per-instance restrictions, and the trivial convex-hull reformulation that every problem admits.

2. *Impossibility* (Section 3). Theorem A: a lossless uniform convex relaxation forces the feasible boundary-data set to be convex. Theorem B and an explicit ball bound: the 5-DOF feasible set is nonconvex, with a gap of metres per second. Theorem B*: nonconvexity persists with altitude and glide-slope constraints at every horizon, with upright landing, and with free final time on compact windows. A line-search lemma shows that searching over t_f cannot rescue LCvx. Theorem D and Lemma S give per-instance versions for minimum-landing-error problems, with a Wirtinger-type inequality that handles coning manoeuvres. Theorem A-opt and Proposition T give the optimization version. Results for $\rho_1 = 0$ show that the obstruction does not depend on a minimum throttle. Transfers to kinematic and independent-torque 6-DOF models are given, and pure TVC is treated through a centre-of-percussion reduction, with its exact scope stated.

3. *Possibility* (Section 4). LCvx is exact for any fixed attitude profile. For the gimbal-torque-coupled TVC body the problem with a given attitude is exactly convex, with no relaxation. There is an exact subclass for the kinematic model. For degree-2 SDP lifts we give stress certificates and a certified-global coplanar subclass, together with counterexamples to exactness in general.

4. *Numerics* (Section 5). Gap laws, results beyond the proven regimes, and the size of the obstruction for a small lander: centimetre-scale for in-air descents, a metre-scale hole in the landing set for pad-launched hops.

5. *Formalization* (Section 6). Theorem A, Theorem B, the state-constrained Theorem B*, the abstract Theorem D and the line-search lemma are machine-checked in Lean 4 with Mathlib [24, 25]. External theorems used on the positive side (the maximum principle and Aumann's theorem) enter only as explicit hypotheses.

Scope, honestly stated. The impossibility concerns the class of convex programs in which the physical state is a decision variable and the boundary data enter affinely. That covers every LCvx formulation in the literature. It does not cover convex programs over measures with point-mass data entry, which are exact but infinite-dimensional and per instance, and it does not say that 5-DOF guidance is intractable. For the in-air descents of the motivating vehicle, the nonconvexity we can exhibit is real but thin. For the actual pure-TVC actuation with an upright terminal attitude, the rigorous proof is still being completed (Section 3.8.1).

Notation. $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ are Euclidean, S^2 is the unit sphere, e_z points up, $x_z = \langle x, e_z\rangle$ and $x_h = x - x_z e_z$. For unit vectors, $\angle(x,y) = \arccos\langle x, y\rangle \in [0,\pi]$. Gravity is $g = -g_0 e_z$ unless stated otherwise. All functions live on $[0, t_f]$ with Lebesgue measure.

2 Problem formulation

2.1 Translational dynamics and thrust

All models share the translational part

$$\dot r = v, \qquad \dot v = a(t)\,d(t) + g, \qquad \dot m = -\alpha T, \qquad a = T/m, \tag{1}$$

with thrust magnitude $T(t) \in [\rho_1, \rho_2]$ (measurable), $0 < \rho_1 \le \rho_2$ unless $\rho_1 = 0$ is stated, mass-depletion constant $\alpha \ge 0$, initial mass m_0 with $m_0 - \alpha\rho_2 t_f > 0$, and thrust direction $d(t) \in S^2$ (measurable). Integrating (1) gives

$$v(t_f) = v_0 + g t_f + I, \quad r(t_f) = r_0 + v_0 t_f + \tfrac{1}{2} g t_f^2 + J, \qquad I = \int_0^{t_f} a\,d\,ds, \quad J = \int_0^{t_f} (t_f - s)\,a\,d\,ds. \tag{2}$$

Since $T \geq \rho_1$ forces $m(s) \leq m_0 - \alpha\rho_1 s$, every admissible acceleration satisfies $a(s) \geq a_1(s) := \rho_1/(m_0 - \alpha\rho_1 s)$, the acceleration at constant minimum throttle.

2.2 Attitude models

The body (thrust) axis is $b(t) \in S^2$. The gimbal constraint is $\angle(d(t), b(t)) \leq \delta$ with $\delta \in [0, \pi/2)$. A body-fixed box gimbal $\pm\delta$ per axis is handled by replacing δ by the corner angle δ^* in obstruction arguments ($\delta^* \approx 21.2°$ for $\pm 15°$) and by the inscribed angle where a witness needs it [`math/adversary-6dof-report.md, V7`].

M0 (3-DOF LCvx baseline).
No attitude; optional pointing cone $\angle(d, \hat{n}) \leq \theta_p$ about a fixed axis.

M1 (5-DOF kinematic: rate-limited axis, gimbal cone).
b is continuous, $b(0) = e_z$, and *angle-Lipschitz*:

$$\langle b(s), b(s') \rangle \;\geq\; \cos(\omega|s - s'|) \qquad \text{whenever } \omega|s - s'| \leq \pi. \tag{AL$_\omega$}$$

(AL$_\omega$) holds iff b is absolutely continuous with $\dot{b} = \Omega \times b$ and $\|\Omega_\perp\| \leq \omega$, where $\Omega_\perp$ is the transverse body rate; roll is unrestricted [`math/02, Prop. 1.2`]. The gimbal is fast (d measurable). Two terminal variants: (F) $b(t_f)$ free; (T) $b(t_f) = e_z$.

M3a / M4-dyn (rigid body, independent transverse torque).
Axisymmetric inertia $J = \mathrm{diag}(J_t, J_t, J_a)$, Euler's equations, transverse torque $\|\tau_\perp\| \leq \tau_t$ from an actuator independent of the main engine (reaction-control system or wheels), any roll torque, $\Omega(0) = 0$. Optional rate limit.

M4-kin (kinematic 6-DOF).
Full attitude $\mathsf{R}(t) \in SO(3)$ with angular velocity as control, $\|\Omega_\perp\| \leq \omega$, arbitrary roll; thrust axis $b = \mathsf{R}e_3$.

PT / M3b (pure TVC rigid body; the realistic small-lander model).
Pitch and yaw torque come only from the gimbal: with the pivot at $-Lb$ from the centre of mass, $\tau_\perp = -LT\, b \times d$. Deflecting the thrust outward rotates the body *inward*. Roll is held by a separate RCS.

Roll does not move the thrust axis of an axisymmetric vehicle. 5-DOF (translation plus pitch and yaw of b) is therefore the natural primary model, and 6-DOF results follow by projection and lifting (Section 3.8).

Lemma 2.1 (Centre-of-percussion identity [`math/02, Lemma 6.4`]). *Consider model PT with constant mass m, axisymmetric J, roll rate $\varpi = \langle \Omega, b \rangle$ and $\varepsilon := J_t/(mL)$. The point $y = r + \varepsilon b$ satisfies*

$$\ddot{y} = \Big(\frac{T\cos\mu}{m} - \varepsilon\|\Omega_\perp\|^2\Big)b + g + \frac{J_a}{mL}\,\varpi\,\Omega_\perp, \qquad \mu = \angle(d, b).$$

With zero roll rate the lateral gimbal force cancels exactly: y is driven by a thrust along b (effective gimbal 0) whose magnitude is reduced by the centripetal term. *[proved in project notes; registry status CLAIMED]*

For the motivating vehicle, $\varepsilon \approx 1.35$ m. The identity is the basis of both the pure-TVC obstruction (Section 3.8.1) and the pure-TVC positive result (Theorem 4.4).

2.3 Boundary data and feasible sets

The boundary data are $\theta = (r_0, v_0) \in \mathbb{R}^6$. The mass m_0, the initial attitude (and initial rate in torque models), the terminal attitude in variant (T), and t_f are fixed. Landing means $r(t_f) = 0$, $v(t_f) = 0$.

Definition 2.2 (Feasible boundary-data sets)**.** $F_{t_f} = \{\theta : \text{some admissible control lands from } \theta \text{ at } t_f\}$. For a closed state-constraint set $\mathcal{X} \subseteq \mathbb{R}^6$, $F_{t_f}^{\mathcal{X}}$ is the subset whose landing trajectory satisfies $(r(t), v(t)) \in \mathcal{X}$ for all t. Our main choice is altitude plus glide slope,

$$\mathcal{X} = \{(r, v) : \ r_z \geq 0, \ r_z \geq \tan\gamma_{gs}\|r_h\|\}, \qquad \gamma_{gs} \in [0, \pi/2). \tag{3}$$

The *velocity-increment set* is $R = \{I\}$ over admissible controls, and $K = \{(I, J)\}$.

By (2), $F_{t_f} = A^{-1}(K)$ for an affine bijection A, and $\pi_v(F_{t_f}) = -gt_f - R$. Hence

$$F_{t_f} \text{ convex} \iff K \text{ convex} \implies R \text{ convex}.$$

For one fixed instance θ_0 we also use $\mathrm{Reach}(\theta_0)$, the set of terminal states of admissible trajectories that stay in $\mathcal{X}$, and the *landing-offset set* $L(\theta_0) = \{p : \text{some admissible trajectory from } \theta_0 \text{ ends at } r(t_f) = (p, 0), \ v(t_f) = 0\}$.

2.4 Relaxations and losslessness

Definition 2.3 (Uniform convex relaxation)**.** Let $\Theta = \mathbb{R}^6$ and let Y be any real vector space (lifted variables, discretized controls, slacks; possibly infinite-dimensional). A *uniform convex relaxation* is a convex set $C \subseteq \Theta \times Y$. Its slice $C_\theta = \{y : (\theta, y) \in C\}$ is the feasible set of the relaxed problem at θ, which may minimize any cost. The data enter *affinely* in the sense that C is convex jointly in (θ, y). More generally, they enter through an affine bijection ψ of Θ, or through a *separable monotone* map $\psi(\theta) = (\psi_r(r_0), \psi_1(v_{0x}), \psi_2(v_{0y}), \psi_3(v_{0z}))$ with each ψ_i continuous and strictly monotone in a frame with vertical third axis.

Definition 2.4 (Losslessness)**.** (a) *Feasibility (detection-sound) form:* C is lossless for $F \subseteq \Theta$ if $C_\theta \neq \emptyset \iff \theta \in F$ for every θ. It is *ε-lossless* if $F \subseteq \pi_\Theta(C) \subseteq \{\theta : \mathrm{dist}(\theta, F) \leq \varepsilon\}$. (b) *Optimization (LCvx) form* for a cost J: every original-feasible θ has $C_\theta \neq \emptyset$, and whenever $C_\theta \neq \emptyset$ the relaxed minimum over C_θ is attained at a point that certifies an original-feasible (and optimal) solution at θ. (c) *Value-lossless on F:* $\min_{C_\theta} J = V(\theta)$ for every $\theta \in F$, where V is the original value.

Proposition 2.5 (LCvx form implies feasibility form [`math/02, Prop. 1.5`])**.** *If C is lossless in the optimization form (b) for some cost, it is lossless in the feasibility form (a).* *[Lean-checked]*

Proof. The first half of (b) gives "$\Leftarrow$". If $C_\theta \neq \emptyset$, the attained relaxed optimum certifies an original-feasible solution, so $\theta \in F$. □

So the feasibility form is not stronger than the definition used in the LCvx literature [4, 14], provided relaxed optima are attained (which thrust bounds guarantee). A relaxation that is exact on F but reports false feasibility off F fails (b). That is the failure mode Theorem A forces, and the optimization form (c) is treated separately in Section 3.6.

Definition 2.6 (State-linear relaxation of one instance [math/02, Def. 11.1])**.** Fix an instance θ_0. A *state-linear relaxation* is (Y, C, ρ, P) with $C \subseteq Y$ convex, $\rho : Y \to \mathbb{R}^6$ affine (the physical readout of the terminal state, or of (r, v) at finitely many times), and $P : Y \to [0, \infty]$ convex (a penalty), such that every genuine trajectory with terminal state x_f is represented by some $y \in C$ with $\rho(y) = x_f$ and $P(y) = 0$. For a convex terminal-error function φ, the relaxed problem is $\inf_{y \in C} \varphi(\rho(y)) + P(y)$; the original is $\inf_{x \in \mathrm{Reach}(\theta_0)} \varphi(x)$.

2.5 Why this class captures LCvx, and what it excludes

Every LCvx formulation we know enters (r_0, v_0) through linear equality constraints $x(0) = x_0$ on top of a convex constraint set in the remaining variables (states, relaxed controls, slack Γ, discretization). It is therefore a uniform convex relaxation at fixed t_f. The log-mass change of variables $z = \ln m$ [2] acts on Y, and the only data reparametrization in the literature is coordinatewise monotone. The cost of LCvx (fuel, landing error) is a convex function of the physical variables plus slack penalties that vanish on genuine points, which is Theorem 2.6. Statements are made in continuous time. Discrete-time LCvx is not exact in general either [12], and continuous-time statements avoid conflating the two effects.

The class excludes the following on purpose. We consider these the honest boundary of the impossibility.

1. *Nonaffine data entry.* Any compact $S \subset \Theta$ is the preimage of a convex set of measures under $\theta \mapsto \delta_\theta$ (point mass). Without a restriction on how data enter, convexity of a feasible set has no content.
2. *Occupation-measure and moment-SOS lifts* [26, 27]. Here the initial state enters as a point mass, terminal conditions act on the support of a lifted measure, and costs are integrals against it. These programs are convex and, under standard conditions, value-exact [28, 29], but they are infinite-dimensional and set up per instance. Their order-one truncation is exactly the affine case and inherits the obstruction. Degree-2 truncations are studied in Section 4.5.
3. *Costs on lifted second moments* (e.g. $R \succeq r(t_f) r(t_f)^\top$ with cost $\mathrm{tr}\, R - 2q^\top r$). On mixtures these average the error instead of evaluating it at the mean readout.
4. *Per-instance restrictions* such as a fixed attitude profile. These violate the embedding property of Theorem 2.6, and they are the positive side of the story (Section 4).

Proposition 2.7 (Trivial exact convex reformulation [math/02, Prop. 11.3])**.** *Every optimal control problem* $\min_{x \in \mathcal{F}} J(x)$ *with* J *bounded below has an exact convex reformulation: minimize the linear functional* $(x, t) \mapsto t$ *over* $\mathrm{conv}\{(x, J(x)) : x \in \mathcal{F}\}$*. If the graph is compact, the minimum is attained at a genuine solution.* *[proved in project notes; registry status CLAIMED]*

This reformulation is per instance, not state-linear (the cost is a lifted coordinate that averages over mixtures), and describing the hull is as hard as the original problem. "Is LCvx possible?" is therefore a meaningful question only inside a tractable class. Our answer is the statement that the state-linear class cannot be exact.

3 Impossibility results

3.1 The abstract obstruction

Theorem 3.1 (Theorem A)**.** *If a uniform convex relaxation* C *with affine data entry is lossless for* F *(feasibility form), then* $F = \pi_\Theta(C)$*; in particular* F *is convex. Consequently, if* F *is not convex,*

no uniform convex relaxation is lossless for F, in either the feasibility or the optimization form, whatever the cost. [Lean-checked]

Proof. $\theta \in \pi_\Theta(C) \iff C_\theta \neq \emptyset \iff \theta \in F$, and a linear image of a convex set is convex. The optimization form reduces to the feasibility form by Theorem 2.5. □

Corollary 3.2. *(i) If the data enter through an affine bijection ψ, then $\psi(F)$, hence F, is convex. (ii) If C contains the original problem ($F \subseteq \pi_\Theta(C)$), then $\pi_\Theta(C) \supseteq \operatorname{conv} F$, so C reports feasibility on all of $\operatorname{conv} F \setminus F$. If $\operatorname{dist}(\theta_M, F) = \Gamma > 0$ for some $\theta_M \in \operatorname{conv} F$, no uniform convex relaxation is ε-lossless for $\varepsilon < \Gamma$.* [Lean-checked] *(part (i))*

Theorem A is elementary. The substance lies in showing that the 5-DOF and 6-DOF feasible sets are not convex, in the physically relevant settings, and in establishing that the definitions match LCvx (Section 2.5). Theorem A also explains why 3-DOF LCvx is consistent: for the exact 3-DOF problem F_{t_f} is convex by Aumann's theorem [23] (Theorem 3.22).

Lemma 3.3 (Reduction to velocity increments `[math/02, Lemma 3.1]`)**.** *Let $P_\pm \in R$ be realized by controls landing from $\theta_\pm$, $M = \frac{1}{2}(P_+ + P_-)$ and $\theta_M = \frac{1}{2}(\theta_+ + \theta_-) \in \operatorname{conv} F_{t_f}$. Then $\operatorname{dist}(\theta_M, F_{t_f}) \geq \operatorname{dist}(M, R)$. In particular $M \notin R \Rightarrow \theta_M \notin F_{t_f}$.* [Lean-checked]

3.2 Theorem B: the unconstrained 5-DOF feasible set is nonconvex

Let $\varphi_F(s) = \omega s$ (variant F) or $\varphi_T(s) = \omega \min(s, t_f - s)$ (variant T). For a horizontal unit vector u, the *extreme arc* tilts the body at full rate toward u, deflects the thrust a further δ outward, and holds minimum throttle:

$$b_u = \cos\varphi\, e_z + \sin\varphi\, u, \quad d_u = \cos(\delta+\varphi)\, e_z + \sin(\delta+\varphi)\, u, \quad T \equiv \rho_1. \tag{4}$$

Its velocity increment is $P_u = (\int a_1 \sin(\delta+\varphi))u + (\int a_1 \cos(\delta+\varphi))e_z$.

Theorem 3.4 (Theorem B)**.** *Model M1 with $\rho_1 > 0$, $0 < \omega < \infty$, $\delta \in [0, \pi/2)$, $\alpha \geq 0$, any g, and either (F) $b(t_f)$ free with $\delta + \omega t_f \leq \pi/2$, or (T) $b(t_f) = e_z$ with $\delta + \omega t_f/2 \leq \pi/2$. Then $M = \frac{1}{2}(P_u + P_{-u}) = (\int_0^{t_f} a_1 \cos(\delta+\varphi))\, e_z$ lies in $\operatorname{conv} R \setminus R$. Hence R and F_{t_f} are nonconvex, and no uniform convex relaxation of the 5-DOF landing problem is lossless.* [Lean-checked: variant F, any continuous $a_1 > 0$] [proved in project notes; registry status CLAIMED] *(variant T)*

Proof sketch (full proof in Section A). Suppose $M = \int a\, d$ for an admissible control. (1) By (AL_ω) from $s' = 0$, $b_z(s) \geq \cos\varphi(s)$. The spherical triangle inequality and the gimbal constraint give $d_z \geq \cos(\delta+\varphi) \geq 0$, so $a\, d_z \geq a_1 \cos(\delta+\varphi)$ pointwise. (2) Integrating, $M_z \geq \int a_1 \cos(\delta+\varphi) = M_z$, so equality holds a.e. (3) Equality in the triangle inequality forces $b_z = \cos\varphi$ and puts d in the vertical plane of b, tilted outward: $d = \cos(\delta+\varphi)e_z + \sin(\delta+\varphi)u(s)$ with $u(s)$ the azimuth of $b(s)$. (4) *Rigidity:* for $s < s'$ on a monotone piece of φ, (AL_ω) gives $\langle b(s), b(s')\rangle \geq \cos(\varphi(s') - \varphi(s))$, which with both tilts exact forces $u(s) = u(s')$. So the azimuth is constant. (5) Then $M_h = (\int a \sin(\delta+\varphi))u_0 \neq 0$, a contradiction. □

Step (4) is where the finite rate enters. With $\omega = \infty$ the azimuth can chatter between $\pm u$ and $M \in R$, which is consistent with 3-DOF LCvx. With $\omega = 0$ the attitude is fixed and $M \in R$ again, by chattering of the thrust azimuth on the gimbal cone. So $\omega \in (0, \infty)$ is essential. The gimbal is not: $\delta = 0$ works.

Quantitative gap: the ball bound

Theorem 3.5 (Theorem Q, ball bound [math/02, §5]). *In M1, variant F, assume $\delta + \omega t_f/2 \le \pi/2$ and $M_z = \int a_1 \cos(\delta + \omega s)\, ds \ge 0$ (automatic if $\alpha = 0$ or $\delta + \omega t_f \le \pi/2$). Every $I \in R$ satisfies $\|I\| \ge \langle I, b(t_f/2)\rangle \ge G(t_f/2)$, where $G(\tau) = \int_0^{t_f} a_1 \cos(\delta + \omega|s - \tau|)\, ds$. Consequently*

$$\operatorname{dist}(\theta_M, F_{t_f}) \;\ge\; \operatorname{dist}(M, R) \;\ge\; G(t_f/2) - M_z \;\ge\; \Gamma_F := \frac{4a_1(t_f/4)}{\omega} \sin\Big(\delta + \frac{\omega t_f}{2}\Big) \sin^2\Big(\frac{\omega t_f}{4}\Big) > 0,$$

with equality in the last step if $\alpha = 0$, and no uniform convex relaxation is ε-lossless for $\varepsilon < \Gamma_F$. [proved in project notes; registry status CLAIMED]

The proof needs only the triangle inequality, without its equality case, and one explicit integral (Section A). For $\rho_1 = 400$ N, $m_0 = 80$ kg, $\delta = 15°$, $\omega = 0.3$ rad/s, $t_f = 5$ s the bound is 7.6 m/s. Numerically it is within 17–19% of the true distance over 39 configurations (Section 5). For small ωt_f, $\Gamma_F \approx \frac{1}{4} a_1 \omega t_f^2 \sin\delta$ ($\delta > 0$), so the gap vanishes as $\omega \to 0$ and is linear in ω. A two-reference-direction refinement gives an explicit gap $\gamma_T > 0$ for the tent variant (T) when $\alpha = 0$ [math/02, Thm Q_T].

Proposition 3.6 (All horizons, free terminal attitude [math/02, Thm L]). *In M1 with $\rho_1 > 0$, $0 < \omega < \infty$, $0 \le \delta < \pi$, $\alpha \ge 0$ and $b(t_f)$ free, F_{t_f} is nonconvex for* every *$t_f > 0$. With $\rho_1 = 0$ and $\alpha = 0$ the same holds whenever $\delta + \omega t_f > \pi/2$.* [proved in project notes; registry status CLAIMED]

The witnesses of Theorem 3.6 use capped arcs (and inverted thrust when $\delta + \omega s > \pi/2$), so their physical content lies mostly in Theorems 3.4 and 3.5. They do show that the obstruction is not an artefact of short horizons.

3.3 State constraints: the dive-then-brake face (Theorem B*)

The witnesses of Theorems B and Q fly at minimum throttle throughout. A small lander cannot hover at minimum throttle ($\rho_1/(m_0 g_0) \approx 0.51$ for our vehicle), so these witnesses start *below ground*. Theorems B and Q say nothing about the altitude-constrained problem. The repair uses a different exposed face. Fix a switch time $s^* \in (0, t_f)$ and the rotation-invariant linear functional $\ell(\theta) = r_{0z} + s^* v_{0z}$. A landing control satisfies the exact identity

$$\Lambda(\theta) := r_0 + s^* v_0 = \int_0^{t_f} (s - s^*)\, a\, d\, ds + g_0 t_f (s^* - \tfrac{t_f}{2}) e_z. \tag{5}$$

The weight $s - s^*$ is negative before s^* and positive after. Maximizing ℓ therefore asks for a *dive* (minimum throttle, tilted as far as possible) followed by a *brake* (full throttle, vertical thrust). The brake keeps the trajectory above ground.

Theorem 3.7 (Theorem B*). *Model M1 with $\rho_1 > 0$, $0 < \omega < \infty$, $0 < \delta < \pi/2$, $\alpha \ge 0$, $A_b := \rho_2/m_0 - g_0 > 0$, and the state constraint (3). Let $s^* \in (0, t_f)$ satisfy*

$$\textit{(S1)}\ \omega s^* \le \delta, \qquad \textit{(S2)}\ g_0 s^* < A_b(t_f - s^*), \qquad \textit{(S3)}\ \tan\gamma_{gs}\, a_{\max} s^{*2} \le A_b (t_f - s^*)^2.$$

Then there are $\theta^* \in F_{t_f}^{\mathcal{X}}$ and its rotation $R_\pi \theta^* \in F_{t_f}^{\mathcal{X}}$ by π about e_z whose midpoint θ_m (with zero lateral components) is not in the unconstrained *set F_{t_f}. Hence $F_{t_f}^{\mathcal{X}}$ is nonconvex for every $t_f > 0$ (for every t_f a small enough s^* exists), and no uniform convex relaxation of the altitude- and glide-slope-constrained 5-DOF problem is lossless. The same proof covers: (T′) upright landing $b(t_f) = e_z$ when $t_f \ge 2s^*$; free final time on any compact window $[t_a, t_b]$, i.e. $\bigcup_{t \in [t_a, t_b]} F_t^{\mathcal{X}}$ is nonconvex; any*

additional closed, R_π-invariant constraint satisfied by the witness (e.g. a tilt cap $\geq \omega s^*$). If moreover $\delta \leq \pi/4$, then

$$\operatorname{dist}(\theta_m, F_{t_f}) \;\geq\; \frac{\eta_0}{\sqrt{1+s^{*2}}}, \qquad (1+c_1)\eta_0 + \sqrt{2W_b\eta_0} = \gamma_X,$$

with $\gamma_X = \frac{5}{32}a_{10}\sin\delta\, s^{*2}$, $c_1 = 4a_{\max}/(\omega a_{10}\sin\delta\, s^*)$, $W_b = \frac{1}{2}a_{\max}(t_f - s^*)^2$, $a_{10} = \rho_1/m_0$, $a_{\max} = \rho_2/(m_0 - \alpha\rho_2 t_f)$. [Lean-checked: qualitative statement, $\alpha = 0$, $b(t_f)$ free, every t_f, any $\mathcal{X} \supseteq$ glide-slope set] [proved; survived attack] *(all variants and the quantitative bound)*

Proof sketch (full proof in Section B). For a fixed throttle, $W_z = \int (s - s^*)a\, d_z$ is bounded above pointwise: on the dive $d_z \geq \cos(\delta + \omega s)$ and the weight is negative; on the brake $d_z \leq 1$. One attitude history attains both bounds. It dives at full rate for s^* and then holds its tilt $\omega s^* \leq \delta$, so vertical thrust remains inside the gimbal cone. Full throttle on the brake is optimal by a pointwise mass comparison. The witness pair lands from θ^* and $R_\pi\theta^*$. It stays above ground, because the brake decelerates at $\geq A_b$, and inside the glide slope by (S3). If θ_m were feasible, its control would attain the bound, so the equality case and the rigidity argument of Theorem B apply on the dive and force $d = e_z$ on the brake. Its weighted lateral moment would be nonzero, whereas $\Lambda_h(\theta_m) = 0$. The quantitative part avoids the equality case. It uses a support bound at $\tau = \vartheta s^*$ for the dive and Cauchy–Schwarz for brake-phase tilts. □

Remark 3.8 (Formal version and its hypotheses). The Lean statement `not_convex_constrainedLandingSet` assumes constant acceleration bounds $0 < a_{\min} \leq a \leq a_{\max}$ (constant mass), $g_0 \leq a_{\max}$, $\omega s^* \leq \delta$, $\delta + \omega s^* \leq \pi/2$, and the single smallness condition $(a_{\max} + 2\tan\gamma\, a_{\min})s^{*2} \leq (a_{\max} - g_0)(t_f - s^*)^2$ in place of (S2)–(S3). `exists_switchTime` shows that such an s^* exists for every $t_f > 0$, $0 < \delta < \pi/2$, $g_0 < a_{\max}$. The statement holds for every $\mathcal{X}$ containing the glide-slope set.

Remark 3.9 (The B* gap is thin, and why). For the motivating vehicle ($\rho_1 = 400$ N, $\rho_2 = 1000$ N, $m_0 = 80$ kg, $\delta = 15°$, $\omega = 0.4$ rad/s, $\gamma_{gs} = 30°$, $s^* = \delta/\omega$, $t_f = 7$ s), $\gamma_X = 0.087$ m but $\eta_0 \approx 1.3 \times 10^{-5}$ m. This is not slack in the proof. A competitor can cancel the dive's lateral moment by tilting slightly during the long brake, at a cost quadratic in the tilt, of order $\|X_h\|^2/(2W_b)$. Dropping (S1) gives an envelope version (Theorem B*-env, `[math/02 §10.6]`; survived adversarial review) whose heuristic gap is at most about 1 cm. Numerically the B* face gap is about 1 cm/s (Section 5). *So the nonconvexity of the real constrained problem is proved, but a practically significant ε-impossibility for in-air descents of this vehicle is not.*

3.4 Free final time and line search

LCvx usually treats free final time by a line search over t_f, solving one convex program per horizon [4, 5]. Two routes are possible. A single program in which t_f is a jointly convex decision variable fails immediately: every slice F_t would then be convex `[math/02, Prop. 7.2]`. Line search needs an argument, because the accepted set is a union of convex sets.

Lemma 3.10 (Line search cannot be lossless; RR-LS)**.** *Let W be a set of horizons, F_t the true feasible sets, and $\{C_t\}_{t\in W}$ per-horizon uniform convex relaxations with $F_t \subseteq \pi_\Theta(C_t)$. (i) If some $\bar{x} \in \operatorname{conv} F_{t^*}$, $t^* \in W$, lies in no F_t, $t \in W$, then line search accepts $\bar{x}$ although it is infeasible at every horizon. (ii) Such an $\bar{x}$ exists if there are a linear involution R preserving every F_t and an R-invariant linear functional ℓ with a maximizer over the graph $\{(t, x) : t \in W, x \in F_t\}$, and every maximizer has $x \neq Rx$.* [Lean-checked: abstract (i)–(ii)]

Corollary 3.11. *On any compact window on which the B* conditions hold (e.g. $W \subseteq [t_a, \min(t_b, t_{fuel})]$ with $t_{fuel} = (m_0 - m_{dry})/(\alpha\rho_1)$), the B* functional supplies (ii). No per-horizon*

family of uniform convex relaxations makes free-final-time 5-DOF landing lossless, in feasibility or optimization form. [proved in project notes; registry status CLAIMED]

The window is necessary for this witness: the unconstrained midpoint of Theorem B becomes reachable at a longer horizon when $g_0 > a_1 \cos\delta$ [`math/redteam-lift §4.3`], and over unbounded horizons $\sup \ell = \infty$.

3.5 Per-instance obstructions: Theorem D and Lemma S

Theorem A concerns relaxations that are uniform in the data. A guidance computer solves one instance at a time, so one may ask whether some convex program is exact for each given instance, possibly exploiting a rotated frame or symmetry. Theorem 2.6 answers this for minimum-landing-error problems.

Theorem 3.12 (Theorem D)**.** *For any state-linear relaxation of an instance,* $\rho(\{y \in C : P(y) = 0\}) \supseteq \operatorname{conv}\operatorname{Reach}(\theta_0)$. *Hence the relaxed value is at most* $\inf_{\operatorname{conv}\operatorname{Reach}} \varphi$. *If* $q \in \operatorname{conv}\operatorname{Reach} \setminus \operatorname{cl}\operatorname{Reach}$ *and* $\varphi = \|\cdot - q\|$, *the relaxed value is* 0, *the true value is* $\operatorname{dist}(q, \operatorname{Reach}) > 0$, *and every relaxed optimum reads out a non-genuine state.* [Lean-checked]

Proof. Mix the embedded points y_i of genuine states x_i. The readout is affine and the penalty convex and vanishing at the y_i. □

Theorem D needs no joint convexity in the data and is immune to frame rotation. It is useful only once a nonconvex reachable set is exhibited at a *single* feasible instance. With a soft terminal velocity, the vertical instance θ_m of Theorem B* gives one, with a certified (tiny) gap [`math/02, Cor. D.1`]. The LCvx soft-landing problem imposes $v(t_f) = 0$ hard and penalizes the landing offset. Theorem D then needs a nonconvex landing-offset set:

(S) there is vertical data $\zeta = (r_0 = he_z,\ v_0 = v_{0z}e_z)$ with $0 \notin \operatorname{cl} L(\zeta) \neq \emptyset$.

$L(\zeta)$ is rotation invariant, so (S) says it is an annulus with a hole around the point directly below. The danger is *coning*: precessing the azimuth at fixed tilt cancels lateral motion and might fill the hole. A Wirtinger inequality modulo affine functions handles all such manoeuvres at once.

Lemma 3.13 (Wirtinger modulo affine functions [`math/07, Lemma W`])**.** *Let* P *be the* $L^2(0,T)$*-orthogonal projection onto affine functions. For* $f \in H^1((0,T);\mathbb{R}^N)$,

$$\|f - Pf\|_{L^2}^2 \le \Big(\frac{T}{2\pi}\Big)^2 \Big(\|\dot f\|_{L^2}^2 - \frac{|f(T) - f(0)|^2}{T}\Big),$$

with equality for $f = c(1 - \cos(2\pi s/T))$. *In particular an* ω*-Lipschitz path* β *(of any range dimension) satisfies* $\|\beta - P\beta\| \le \omega T^{3/2}/(2\pi)$. [proved; survived attack]

The inequality is componentwise, so planar S-turns, coning paths and even $\mathbb{R}^N$-valued (lifted) lateral paths obey the same bound.

Theorem 3.14 (Theorem S: Lemma S in an explicit regime [`math/07, Thm S`])**.** *Model M1,* $b(t_f)$ *free,* $\alpha \ge 0$, $g = -g_0 e_z$. *Let* $\bar\tau = \delta + \omega t_f$ *and assume (S0)* $\bar\tau < \pi/2$, *(S1)* $(\rho_1/m_0)\cos(\delta + \omega t_f) > g_0$ *(minimum-throttle thrust-to-weight above* $1/\cos\bar\tau$*; altitude is then automatic), (S2) a throttle condition that is automatic if* $\rho_1 = \rho_2$, *and (S3)* $M := \operatorname{gain}_w - B > 0$, *where* gain_w *is the vertical-impulse waste of an explicit planar single-crossing S-turn witness and* B *is an explicit constant*

(Section C). Then at the witness's vertical data ζ_w, $L(\zeta_w)$ contains a full circle, and every landing offset satisfies

$$|p| \;\geq\; r_S = \Big(\frac{M t_f^3}{12\, C_m}\Big)^{1/2} > 0.$$

Hence (S) holds, and every state-linear relaxation of the minimum-landing-error problem from ζ_w with hard $v(t_f) = 0$ and target $q = 0$ has relaxed value 0 while the true value is at least r_S. [proved; survived attack]

For a demonstration vehicle with $a_1 = 12$, $a_2 = 25$ m/s^2, the theorem is certified for $\omega t_f \leq 0.55$ ($\delta = 0$), ≤ 0.45 ($\delta = 5°$) and ≤ 0.30 ($\delta = 10°$), with certified dents $r_S = 0.4$–1.8 m. *Realism caveat:* (S1) requires a vehicle that cannot hover at minimum throttle. That excludes essentially every deeply throttleable lander, including ours ($\rho_1/m_0 = 5.0$–7.2 m/s$^2 < g_0$). For our vehicle, (S) holds numerically at dive-then-brake data, with landing-set gaps of about 1 cm in ℓ and optimizers landing 1.5–2.4 m off axis (Section 5). The rigorous GTPL case is open, because the L^2 slack (cm–dm) exceeds the ℓ-gap (mm–cm).

3.6 Optimization form: Theorem A-opt and Proposition T

A weaker notion of losslessness escapes Theorem A: exactness of the optimal value on F only, with no infeasibility detection off F. Mixtures of genuine trajectories constrain it too.

Theorem 3.15 (Theorem A-opt [`math/redteam-lift §5`])**.** *Let $C \subseteq \Theta \times Y$ be a uniform convex relaxation with a cost J convex on C, in which every genuine trajectory τ is represented with $J = \text{cost}(\tau)$. If C is value-lossless on F, then $V(\sum_i \lambda_i \theta_i) \leq \sum_i \lambda_i V(\theta_i)$ whenever all θ_i and $\sum \lambda_i \theta_i$ lie in F: the value function agrees with its convex envelope V^{**} on F.* [proved in project notes; registry status CLAIMED]

Proposition 3.16 (Proposition T: an attitude-sensitive cost)**.** *In M1 with $\alpha = 0$, cost $\int_0^{t_f} \langle b, e_z \rangle\, ds$ (maximize mean tilt), $\omega t_f/2 > \delta$, $\delta + \omega t_f/2 \leq \pi/2$ and $a_2 \cos(\omega t_f/2 - \delta) \geq a_1$, there is $\theta' \in F$ at which every such relaxation has value strictly below $V(\theta')$. So no uniform convex relaxation is value-lossless on F for all convex costs.* [proved in project notes; registry status CLAIMED]

For minimum fuel the question is open. Numerically the rotational axis test shows no violation of $V = V^{**}$ [`math/redteam-lift §5.4`]. Even if minimum fuel were value-lossless on F, the relaxation would report spurious feasibility on $\operatorname{conv} F \setminus F$, which is not LCvx in the sense of Theorem 2.4(b).

3.7 The minimum throttle is not the cause: $\rho_1 = 0$

Theorems B and B* use the minimum throttle. If the engine may shut down ($\rho_1 = 0$), the velocity set does become convex in the short-horizon regime, but the landing set does not.

Proposition 3.17 (P0-R [`math/possibility-rho0 §1`])**.** *In M1 with $\rho_1 = 0$, $\alpha = 0$, $b(t_f)$ free: R is convex iff $\delta + \omega t_f \leq \pi/2$, and then $R = \{\int u : \|u(s)\| \leq a_2,\ \angle(u(s), e_z) \leq \delta + \omega s\}$, the 3-DOF time-varying-envelope set. (Tent variant: iff $\delta + \omega t_f/2 \leq \pi/2$.)* [proved in project notes; registry status CLAIMED]

Theorem 3.18 (P0-F: landing set nonconvex with $\rho_1 = 0$)**.** *In M1 with $\rho_1 = 0$, $\alpha = 0$, $b(t_f)$ free, $0 \leq \delta < \pi/4$ and $0 < \omega t_f < \pi/2 - 2\delta$, the set K, hence F_{t_f}, is nonconvex. No uniform convex relaxation is lossless even with engine shutdown allowed.* [proved in project notes; registry status CLAIMED]

Proof idea (Section D). On the face $\langle \hat{p}, r_0 + \sigma v_0 \rangle$ with $\hat{p}$ pointing δ below the horizontal, the face value reduces exactly to a one-dimensional maximization over the tilt reached at σ. An intermediate-value argument in σ produces a tie between two separated groups of maximizers. Both "coast while rotating", and the thrust on/off boundary acts as a convex kink. The face is a disjoint union of two nonempty compact sets, so it is not convex. □

With an altitude constraint the same mechanism gives nonconvexity of $F_{t_f}^{\mathcal{X}}$ (early-kink face) when full-throttle thrust-to-weight is at least $1/(\cos(\delta + \omega t_f)(1 - x_*^2)) \approx 2.1$ [P0-X]. That covers neither our vehicle ($T/W \approx 1.27$) nor glide slopes. The "LCvx is possible iff $\rho_1 = 0$" dichotomy is therefore false: *the obstruction is driven by the finite attitude rate itself.* Theorem 3.17 does give a small exact result: for problems that constrain only the terminal velocity, the 3-DOF envelope SOCP is feasibility-exact when $\rho_1 = 0$.

Two further loopholes are closed. A separable monotone reparametrization of the data cannot convexify F, because the lowest slice of $\pi_v F$ is a circle [math/redteam-lift Prop. P1]. The only rate notion that is SOC-representable (a thrust-vector "chord rate" $\|\dot{u}\| \le \omega\sigma$) is not uniformly lossless, because its original problem is again of type M1 with $\delta = 0$ [math/redteam-reform §6]. A tilt-angle-only rate limit, with the azimuth free to jump, leaves rigidity broken and appears to give convex landing sets numerically (Section 5); it is, however, not SOC-representable.

3.8 6-DOF

Theorem 3.19 (Transfer to kinematic 6-DOF [math/02 Thm 6.2; math/adversary-6dof Thm K])**.** *In M4-kin the projection* $(\mathsf{R}, \Omega, T, d) \mapsto (b = \mathsf{R}e_3, T, d)$ *maps admissible 6-DOF controls into admissible M1 controls, and the planar witnesses lift with zero roll, as rotations about a fixed horizontal axis. Hence* $R_6 \subseteq R_5$ *contains the same witnesses, and Theorems B, Q,* B^* *(and their variants) hold verbatim for 6-DOF with fixed attitude data. With a box gimbal, replace* δ *by the corner angle and take* u *along a diagonal.* [proved in project notes; registry status CLAIMED]

Theorem 3.20 (Independent transverse torque [math/02 Thm E; math/adversary-6dof Thms D, D*])**.** *In M3a/M4-dyn (axisymmetric* J*, transverse torque* $\|\tau_\perp\| \le \tau_t$ *independent of the engine, any roll,* $\Omega(0) = 0$*), every body axis satisfies the envelope* $\angle(b(s), b(s')) \le \Psi(s') - \Psi(s)$ *with* $\Psi(s) = \int_0^s \min(\bar{\alpha} r, \omega)\, dr$*,* $\bar{\alpha} = \tau_t / J_t$*, and the radial "bang" arcs attain it. Hence (i) with terminal attitude free and* $\delta + \Psi(t_f) \le \pi/2$*,* F_{t_f} *is nonconvex; (ii) with terminal attitude* $b(t_f) = e_z$*,* $\Omega_\perp(t_f) = 0$ *and unlimited gimbal rate, a dive-then-brake face again gives nonconvexity, with altitude and glide-slope constraints handled as in Theorem* B^**.* [proved in project notes; registry status CLAIMED]

The gyroscopic terms do no work on $\|\Omega_\perp\|$ for axisymmetric J, so roll and coning cannot defeat the envelope. Not covered are a nonzero initial transverse rate (which breaks the $\pm u$ symmetry), nonaxisymmetric inertia, and gimbal-rate limits in the terminal-attitude case.

3.8.1 Pure TVC: what is proved

On a pure-TVC vehicle the M1 witnesses are not realizable: deflecting the thrust outward rotates the body inward. The centre-of-percussion identity (Theorem 2.1) converts the problem into a zero-gimbal, angular-acceleration-limited problem for $y = r + \varepsilon b$, with boundary data that map affinely. The landing condition becomes a fixed affine condition on $(y, \dot{y})$ only when the vehicle lands upright at rest, $b(t_f) = e_z$, $\Omega_\perp(t_f) = 0$.

Theorem 3.21 (Pure TVC, rest-terminal rate, tilt free [math/adversary-6dof Thm E_0])**.** *Model PT with constant* ε*, zero roll rate, axisymmetric* J*, unlimited gimbal rate (cone or box gimbal),* $b(0) = e_z$*,*

$\Omega_\perp(0) = 0$, *and landing* $r(t_f) = v(t_f) = 0$, $\Omega_\perp(t_f) = 0$ *with terminal tilt* free. *In the regime* $\varepsilon\bar{\omega}^2 < \rho_1 \cos\delta^*/m_0$, $\bar{\omega}t_f \le \pi/2$ *(for a rate cap* $\bar{\omega}$*),* F_{t_f} *is nonconvex (indeed* $\operatorname{cl} F_{t_f}$ *is nonconvex), with no state constraints.* *[proved in project notes; registry status CLAIMED]*

Theorem 3.21 does not cover the realistic case: upright terminal attitude, altitude constraint, and gimbal-rate limits. For that case there is strong numerical evidence (Section 5): a dive-then-brake face with full rigid-body dynamics, ±15° box gimbal at 100°/s, $r_z \ge 0$ and upright-at-rest touchdown has rotation-midpoint gaps of 2.8–3.3 cm at $t_f = 5$–7 s. A torque-coupled hop instance has a 1.74 m hole in its landing set. With $\rho_1 = 0$, pure TVC is nonconvex in a strong-torque, small-gimbal asymptotic regime [P0-TVC].

TODO / placeholder (proof-tvc): Theorem TVC-1, pure TVC upright at rest with $r_z \ge 0$

Status: registry row B6-TVC-up is **OPEN**. `math/06-pure-tvc.md` states Theorem TVC-1: for constant mass, upright at rest at both ends, $r_z \ge 0$ and glide slope, at the GTPL wet constants and certified horizons, $F^{\mathcal{X}}_{t_f}$ is nonconvex. The proof is computer-assisted: an S-procedure/Finsler SDP certificate bounds *every* zero-lateral competitor, is separable in the two components of the horizontal tilt (covering coning and loops), and comes with remainder and localization lemmas. An explicit planar dive-then-brake witness exceeds the bound. The certified-instance table (`CERT_TABLE_PLACEHOLDER`) and the verdict in `comms/reports/proof-tvc.md` are not yet filled in. Adversary attack A1 (additive remainder) is reported fixed. Open: mass depletion, gimbal-rate limit, roll-rate competitors (certification status pending).
To fill in: statement with the certified (t_f, s^*, d_*) table; the scope (constant mass, unlimited gimbal rate, roll R0/R1); the certificate method in one paragraph; an appendix pointer. Until then, the paper claims only Theorem 3.21 and the numerics for pure TVC.

Remark 3.22 (Consistency with 3-DOF). For the exact 3-DOF problem with free direction, F_{t_f} is convex even with mass depletion, and with any measurable compact control set it is convex for constant mass, by Aumann's theorem [`math/02 Prop. 8.2`]. Theorem A therefore never contradicts 3-DOF LCvx in any case where convexity can be decided. The decisive ingredient of our obstruction is that the attitude constraint *couples different times*, through continuity and a rate bound. Pointwise (measurable) attitude constraints never obstruct.

4 What is possible

The impossibility results identify the obstruction precisely: the *choice* of a rate-limited attitude profile. Once that choice is removed, convexification works again, and in the pure-TVC case it works without any relaxation. This section collects the positive results. Aumann's theorem [23] is the one external theorem used without proof. The Pontryagin maximum principle [28, 30] enters only as a labelled hypothesis (H-PMP).

4.1 LCvx is exact for any fixed attitude profile

Fix a measurable $b : [0, t_f] \to S^2$. The original problem $\mathrm{P}_0(b;\theta)$ minimizes $\int \|T\|$ subject to $\rho_1 \le \|T\| \le \rho_2$, $\langle b, T\rangle \ge \|T\| \cos\delta$, the dynamics (1) and the boundary conditions. The relaxation $\mathrm{P}_1(b;\theta)$ minimizes $\int \Gamma$ over $T \in K_\Gamma(b) := \{\|T\| \le \Gamma,\ \langle b, T\rangle \ge \Gamma\cos\delta\}$, $\rho_1 \le \Gamma \le \rho_2$. This is the pointing-cone LCvx of [4] with a time-varying axis.

Theorem 4.1 (Theorem C-W: value and feasibility exactness for any b [math/04 §3]). *For every* P_1*-feasible* (T,Γ) *there is a measurable* $\tilde{T}(t) \in \{\|\tilde{T}\| = \Gamma,\ \angle(\tilde{T},b) \le \delta\}$ *with the same* Γ*, mass history and cost, whose state agrees with that of* (T,Γ) *at* t_f *and at any prescribed finitely many times. Hence* P_0 *and* P_1 *have the same feasible data and the same value, attainment transfers, and nodewise state constraints are preserved. With constant mass (or the convexified-mass model),* $F_{t_f}(b) = \{\theta : \mathrm{P}_0(b;\theta)\ \textit{feasible}\}$ *is convex (Corollary C-F).* [proved in project notes; registry status CLAIMED]

Proof idea. On each subinterval the state increment is linear in T/m for the fixed mass history. The extreme points of the cap $K_\Gamma(b)$ are exactly its spherical part. Aumann's theorem replaces the relaxed selection by a lossless one with the same moments. □

With convex path constraints the value statement holds under a Slater condition. Attainment of P_0 is not claimed in that case.

Theorem 4.2 (Theorem C-S: every relaxed optimum is lossless, under PMP [math/04 §5]). *Assume (H-PMP) for a* P_1*-optimal solution, and let* $p = a + ct$ *be the (affine, for every* b*) primer vector. Off* $D = \{t : p(t) \in -\mathbb{R}_{\ge 0} b(t)\}$ *the optimum is lossless and pointwise unique. With constant mass the lossy set is contained in* $D \cap \{\Gamma^* = \rho_1\}$*. D has positive measure only if* $p \equiv 0$*, or* b *dwells at* $\pm$ *a fixed vector, or* b *follows a* gnomonic line $\pm(a+ct)/\|a+ct\|$ *on a set of positive measure. For analytic* b*,* $\det(b,\dot{b},\ddot{b}) \not\equiv 0$ *suffices for nondegeneracy, and degenerate profiles form a meagre set.* [proved in project notes; registry status CLAIMED]

The result is sharp. For a fixed upright cone and a vertical descent at a suitable altitude, an interior-point solver returns a lossy relaxed optimum with all lossy nodes at minimum throttle. A one-step "purification" SOCP repairs it (Claims C-X, C-K; discrete version: at most three lossy nodes without state constraints, Lean `purification_three`). So *the relaxation itself is never the problem.* This agrees with the lossiness observed in stored GTPL solutions (296 N delivered on a 400 N minimum-throttle arc).

4.2 Per-instance exactness and the union structure

Theorem 4.3 (Theorem C-U [math/04 §9]). $F_{t_f} = \bigcup_b F_{t_f}(b)$ *with each* $F_{t_f}(b)$ *convex (M1 with constant or convexified mass; PT by Theorem 4.4), and* $V(\theta) = \min_b V(b;\theta)$*. In M1 with constant mass the 5-DOF minimum is attained, and the convex program at an optimal attitude* b^* *is exact for that instance.* [proved in project notes; registry status CLAIMED]

The impossibility theorems say exactly that this union of convex sets is not convex. *All* of the nonconvexity lies in the map $\theta \mapsto b^*(\theta)$.

4.3 Pure TVC: exactly convex given the attitude

Theorem 4.4 (Theorem C-6 [math/04 §8]). *In model PT with zero roll rate and constant mass, Euler's equation is equivalent to* $\Pi_b T = -k\Pi_b \ddot{b}$*,* $k = J_\perp/L$*,* $\Pi_b = I - bb^\top$*. For a fixed* $b \in W^{2,\infty}$*, write* $F = -k\Pi_b\ddot{b}$*,* $\varkappa = \|F\|$*. The exact 6-DOF problem is the scalar-throttle program*

$$\min \int \sqrt{T_a^2 + \varkappa^2} \quad s.t. \quad \ddot{r} = (T_a b + F)/m + g, \quad \max\{\varkappa\cot\delta, \sqrt{(\rho_1^2 - \varkappa^2)_+}\} \le T_a \le \sqrt{\rho_2^2 - \varkappa^2},$$

which is convex. No relaxation is involved. Consequently $F_{t_f}^{PT}(b)$ *is convex for every attitude profile. If the body is held upright* ($b \equiv e_z$)*, then* $T \parallel e_z$*: the effective cone is* 0*, so the common "upright body with a* 15° *pointing cone" 3-DOF model is not 6-DOF-feasible for any off-vertical thrust.* [proved in project notes; registry status CLAIMED]

4.4 Exact subclasses of the uniform problem

Proposition 4.5 (X5: small pointing cone, M1 only [`math/redteam-reform §3.1`]). *In M1 with an LCvx pointing constraint $\angle(d, e_z) \le \theta_p$: if $\theta_p \le \delta$ and the initial (and, if prescribed, terminal) attitude lie within $\delta - \theta_p$ of e_z, then $F_{5\mathrm{DOF}} = F_{3\mathrm{DOF}}(\theta_p)$, so classical LCvx guarantees transfer verbatim. If $\theta_p > \delta$ (and the B^* braking condition holds), F_{t_f} is nonconvex. Thus $\theta_p = \delta$ is a sharp threshold* for M1. *[proved in project notes; registry status CLAIMED]*

X5 relies on the gimbal pointing the thrust while the body stays still, which is impossible on a pure-TVC vehicle (Theorem 4.4). There the effective exact class collapses to $\theta_p = 0$. More precisely, a prescribed planar thrust history with $\|T\| \ge \rho_1$ is exactly TVC-realizable only for in-plane initial body rates in an interval of length at most $2\delta\mu_c/\sinh(\mu_c t_f)$, $\mu_c^2 = L\rho_1 \cos\delta / J_\perp$. For our vehicle this is about $3.5\,\mu$rad/s, so a 3-DOF LCvx optimum is essentially never exactly 6-DOF-optimal (Claim C-Y, [proved in project notes; registry status CLAIMED]). In M1, by contrast, $V_5 = V_3$ iff some 3-DOF optimum is δ-trackable by an admissible attitude, with a closed-form primer-arc certificate (Claim C-3D).

4.5 Semidefinite lifts

Occupation-measure linear programs with point-mass data entry are exact but infinite-dimensional and per instance (Section 2.5). Their order-one truncation is exactly the affine case of Theorem A. The first nontrivial tractable truncation is the degree-2 (Shor) lift [31, 32]. We study it for the fixed-thrust minimum-landing-error problem. With weights c_k^v, c_k^r, unknown unit vectors b_k (axis) and d_k (thrust direction), inner-product constraints $\langle d_k, b_k\rangle, \langle d_k, b_{k+1}\rangle \ge \cos\delta$ (gimbal) and $\langle b_k, b_{k+1}\rangle \ge \cos\omega h$ (rate), and cost $\|\sum c_k^v d_k - w_v\|^2 + \|\sum c_k^r d_k - w_r\|^2$, the problem is a vector QCQP. The data w enter only the objective, and the lifted cost is a second-moment cost. The lift therefore lies outside the classes of Theorems A and D, and no contradiction arises.

Lemma 4.6 (Shor lift = Hilbert-sphere problem [`math/08 Lemma 1.1`]). *The Shor relaxation's value equals the minimum of the same vector QCQP posed in $\mathbb{R}^{3+m}$ (indeed in any real Hilbert space containing $\mathbb{R}^3$). The lift is exact iff some optimum can be "planarized" into $\mathbb{R}^3$.* *[proved in project notes; registry status CLAIMED]*

Theorem 4.7 (Stress certificate and local-to-global [`math/08 Thms 2.1, 2.4`]). *Let z^* be a genuine KKT point with multipliers (λ, μ) and stress matrix $Q_\lambda = C + \mathrm{diag}(\mu) - \frac{1}{2}\sum_e \lambda_e (E_{ij} + E_{ji})$. If $Q_\lambda \succeq 0$, then z^* is a global optimum of the lift in every dimension, hence of the genuine problem. Conversely, exactness implies that a PSD stress exists, and $Q_\lambda \succ 0$ gives a unique rank-3 optimum. Moreover the lift is exact iff some genuine global minimizer is a* local *minimizer of the problem posed in $\mathbb{R}^4$ ("$\Leftarrow$" under LICQ). Any LICQ KKT point that is a local minimizer in $\mathbb{R}^4$ is a global minimizer of the lift.* *[proved in project notes; registry status CLAIMED]*

Theorem 4.8 (Coplanar data: certified global optima [`math/08 Thm 4.1`]). *If the lateral parts of w_v and w_r are parallel (velocity-only, position-only or vertical-plane targets), then every planar genuine local minimizer satisfying LICQ is a certified global optimum of both the lift and the genuine problem. This remains true with altitude and glide-slope constraints, or any constraint quadratic in node positions and invariant under rotations about e_z.* *[proved in project notes; registry status CLAIMED]*

Where exactness fails.

- *Unit-thrust discretization with gimbal (numerical).* With $\|d_k\| = 1$ and $\delta > 0$ the lift is lossy on an asymmetric near-dent target: SDP 0.027529 (rank 4) against best genuine 0.028231 over

480 multistarts, a 2.5% gap. Three independent implementations agree, and an explicit feasible $\mathbb{R}^4$ chain attains the SDP value. Gaps are frequent at small N and small ωt_f. The mechanism is identified exactly: in the lift, d_k can tilt into the extra dimension at no rate cost, which shortens its in-plane part, i.e. it throttles one segment down [`math/08 Prop. 6.1`]. A unit-norm discretization forbids this chattering, while continuous time allows it. With the convexified lens ($\|d_k\| \le 1$, the discrete analogue of relaxed controls) the same instance is exact. [numerical evidence only] (the 3-DOF optimum is not certified)

- *Variable throttle (certified).* The natural degree-2 lift with $\|U_k\|^2 = a_k^2$ is lossy whenever the rate limit binds: throttle headroom lets the thrust tilt out of the physical space while keeping its in-plane part. At $N = 3$ an order-2 Lasserre bound, which is a valid lower bound on the genuine problem, equals the local optimum 0.6047, while the Shor value is 0.5759. [certified numerically by Lasserre-2, up to solver accuracy]
- $\delta = 0$ *(no gimbal).* No gap on about 2,900 targets. [numerical evidence only]

TODO / placeholder (verifier-sdp / proof-sdp): Conjecture CSD2′

Conjecture CSD2′ (open; no counterexample known). The Shor lift is exact (a) in continuous time with a fixed thrust profile, (b) for the convexified-lens discretization $\|d_k\| \le 1$, and (c) for the $\delta = 0$ chain. By Theorem 4.7 this is equivalent to the absence of second-order descent directions out of the physical space at genuine optima. Evidence: case 12 after convexification, its coplanar variant, all converged cases of the convexified scan, and 840 adversarial $\delta = 0$ instances. *To fill in:* the outcome of `numerics/verifier-sdp/run_csd2p.py` (convexified check, including the $\delta = 40°$, $N = 4$ instance), and either a proof or a counterexample.

Practical reading. The degree-2 lift is a polynomial-size SDP whose value is always a rigorous lower bound, and a positive value certifies that exact landing is infeasible. Exactness is certified a posteriori by a rank-3 optimum, by the stress test of Theorem 4.7, or a priori in the coplanar case of Theorem 4.8. In the tests, it was exact on every realistic in-air target for our vehicle. It is a certificate, not a lossless convexification. It does not handle minimum fuel: fuel is constant for fixed thrust, and the natural variable-throttle lift is lossy.

5 Numerical evidence

Everything in this section is numerical evidence and is not used in any proof.

Method. Discretized nonlinear programs were solved by multistart interior point (CasADi/IPOPT). For M1 the discretization is *inner*: b is piecewise geodesic with node-to-node angle at most ωh, and d_k must lie within δ of both b_k and b_{k+1}. Spherical caps are geodesically convex, so every discrete solution is an admissible continuous-time trajectory, and every reported distance to a reachable set is an *upper* bound on the true one. Multistart cannot certify global optima, and "no gap found" is weak evidence. Models: M1 (planar and 3-D, with coning initial guesses), M1A (angular-acceleration-limited axis), M3 (torque-coupled TVC rigid body, quaternion attitude, gimbal rate 100°/s), and MT (tilt-angle-rate only, azimuth free). Parameters are those of the motivating vehicle: $m_0 = 80$ kg, $T \in [400, 1000]$ N, $\alpha = 5.66 \times 10^{-4}$ s/m, $\delta = 15°$, $J_\perp = 63.6\ \mathrm{kg\,m^2}$, $L = 0.589$ m, and $\omega = 0.4$ rad/s in the state-constrained studies. A Mars-like parameter set after [2] was also used.

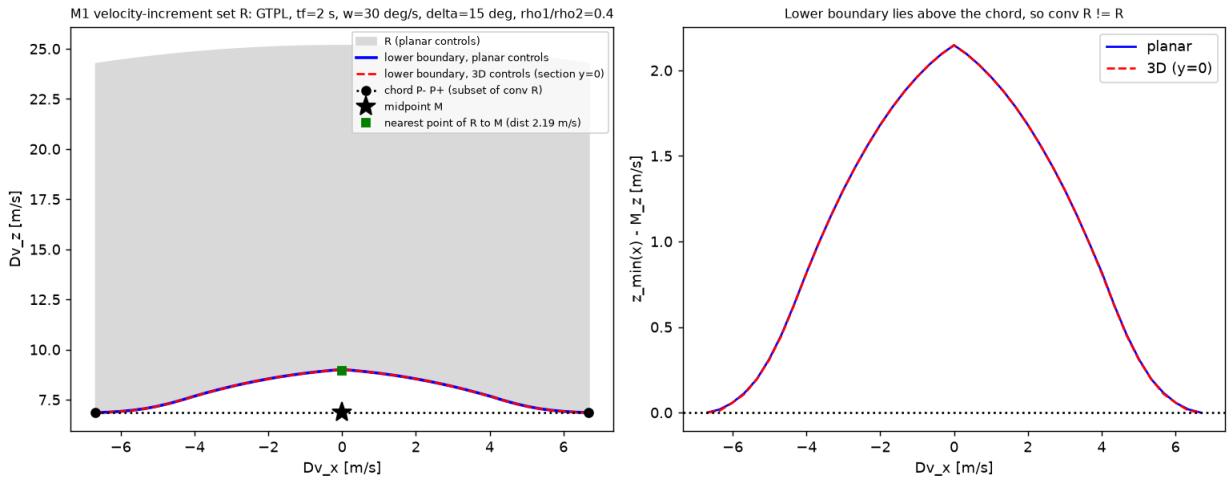


Figure 1: Velocity-increment set R of model M1 ($t_f = 2$ s, $\omega = 30°/\text{s}$, $\delta = 15°$, $\rho_1/\rho_2 = 0.4$). The lower boundary lies strictly above the chord $P_{-u}P_u$, so $M \in \operatorname{conv} R \setminus R$ (Theorem 3.4). Planar and 3-D controls give the same boundary. Numerical (inner discretization).

5.1 Theorem B regime: gap law and the ball bound

For $t_f = 2$ s, $\omega = 30°/\text{s}$, $\delta = 15°$, the midpoint M lies 2.18–2.23 m/s from R, converging in N (Fig. 1). 3-D controls gain essentially nothing over planar ones. Over 39 configurations (ω 10–120°/s, ωt_f 15–75°, δ 2–30°, ρ_1/ρ_2 0.05–1, α $0–5 \times 10^{-3}$) the empirical law is

$$\operatorname{dist}(M, R)/|P_{u,x}| \;\approx\; 0.31\,\omega t_f,$$

nearly independent of δ, ρ_1/ρ_2 and α, and linear in ρ_1 (Fig. 2). The rigorous ball bound of Theorem 3.5 is 81–83% of the NLP upper bound in all 39 cases, with no violation.

5.2 Beyond the proven regimes (no state constraints)

For long horizons without a tilt cap the generalized midpoint gap grows (2.1/13.5/18.6 m/s at $t_f = 2/4/8$ s), but that regime uses inverted flight. With a 30° tilt cap it is 1.15/1.56/0.35 m/s, shrinking as azimuthal sliding along the cap weakens rigidity. In the tent variant the gaps are 0.66 m/s ($t_f = 2$) and 6.24 m/s ($t_f = 5$). For free final time the Theorem B witness, which starts underground, becomes reachable at a longer horizon (Fig. 3). This is why the free-final-time results need B* and a window. The torque-coupled TVC model M3 has nonconvex R with gaps of 0.8–8 m/s at $t_f = 4$ s, and every minimizer of Δv_z has nonzero lateral component.

5.3 With altitude constraints: the physically relevant sets

Gaps of physically admissible data, all with $r_z(t) \geq 0$ at every node, are summarized in Table 1.

- *B^* face.* The rotation-midpoint gap is about 0.010 (weighted m/s) at $t_f = 4$ s and 0.007 at $t_f = 8$ s, stable in N. That is real but tiny, roughly 0.3–0.5% of the face's lateral velocity, consistent with Theorem 3.9.
- *Slice lemma (S) at dive-then-brake data.* Strict gaps of 0.016 m ($t_f = 4$) and 0.009 m ($t_f = 8$), with optimizers landing 1.5–2.4 m off axis. Coning narrows planar gaps by at most about 30% (Fig. 4).

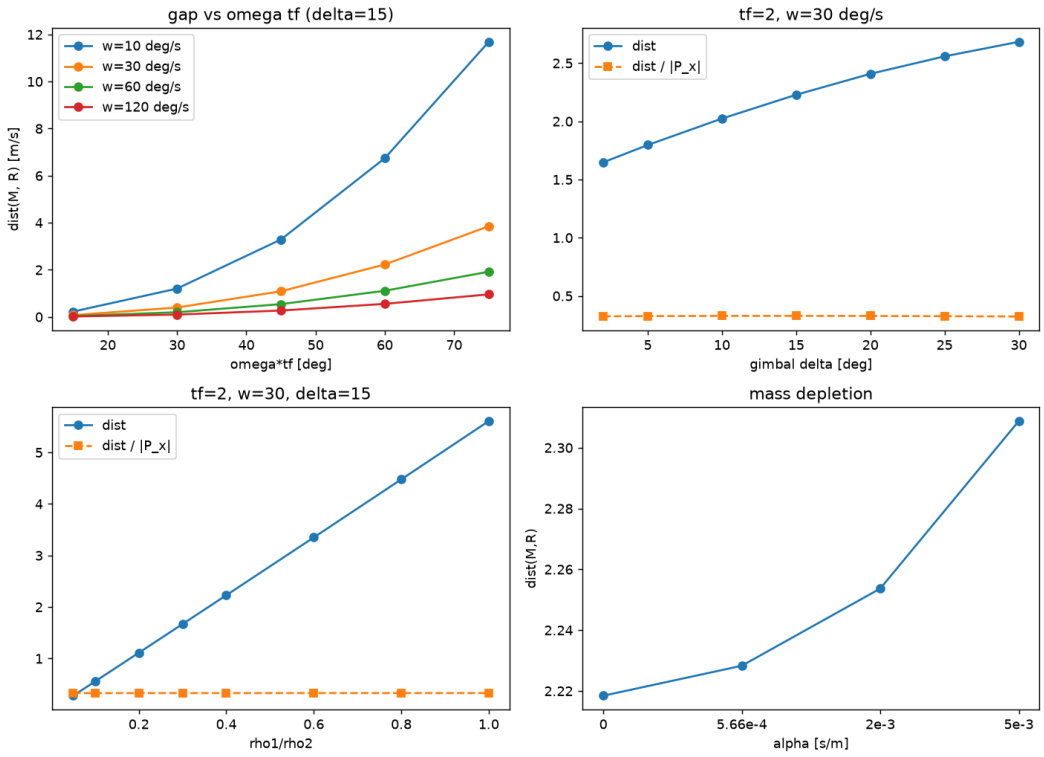


Figure 2: Midpoint gap $\mathrm{dist}(M, R)$ against ωt_f, δ, ρ_1/ρ_2 and α. The normalized gap $\mathrm{dist}/|P_x|$ is flat in δ and ρ_1/ρ_2. Numerical.

- *Face scan and the hop instance.* Among rotation-invariant vertical faces of $F^{\mathcal{X}}$, the pure-dive face is pinned at $r_{0z} = 0$, $v_{0z} > 0$: a vehicle on the pad moving upward. Its midpoint gap is 0.65 ($t_f = 4$) and 2.94 ($t_f = 8$) weighted m/s, 60–250 times the B* gap. At pad data $r_0 = 0$, $v_{0,h} = 0$, $t_f = 4$ s, the landing-offset set L is an annulus whose inner radius grows from 0 to 4.99 m as v_{0z} goes from 7.65 to 7.90 m/s (Fig. 5). Inside the window $v_{0z} \in (7.727, 7.917]$, landing back on the pad is infeasible while landing 0.5–8 m away is feasible. For the torque-coupled TVC model at $v_{0z} = 8.0$ m/s the pad lies 1.74 m from L. The *rigorous* part of the hop result is an exact reduction, an upright pad-landing limit $v_{0z} \le 7.2045$ ($\alpha = 0$) / 7.2833 ($\alpha > 0$), and explicit planar witnesses landing 6.58 m / 4.61 m off the pad at $v_{0z} = 7.7116$ / 7.7425. The *hole* itself is not proved: quadratic and S-procedure certificates are provably too loose here.
- *Tilt-rate-only model MT.* At the same hop data, L is a full disk of radius 10.1 m. The full-direction rate limit, which creates rigidity, is what creates the hole.
- *In-air descent.* From $r_0 = (0, 0, 50)$ m, $v_0 = (4, 0, -5)$ m/s, $t_f = 7$ s, the set L is 104 m wide with no dent above solver noise (≤ 0.016 m).
- *Long horizons and coning.* On the pure-dive face with a 30° tilt cap, the 3-D slice gap falls to 0.69/0.50/0.106 m/s at $t_f = 8/12/20$ s (planar: 0.70/0.80/0.82), and at $t_f = 20$ s the optimizer cones through 359° on the cap. Coning fills about 87% of capped dents, but not all of them.
- *Pure TVC, upright at rest.* Full rigid body, ±15° box gimbal at 100°/s, roll free, $r_z \ge 0$, upright at rest at touchdown, face $\ell = r_{0z} + s^* v_{0z}$ with $s^* = 1$ s: zero-lateral optimum below the free optimum by 2.8×10^{-2} m ($t_f = 5$ s) and 3.3×10^{-2} m ($t_f = 7$ s).

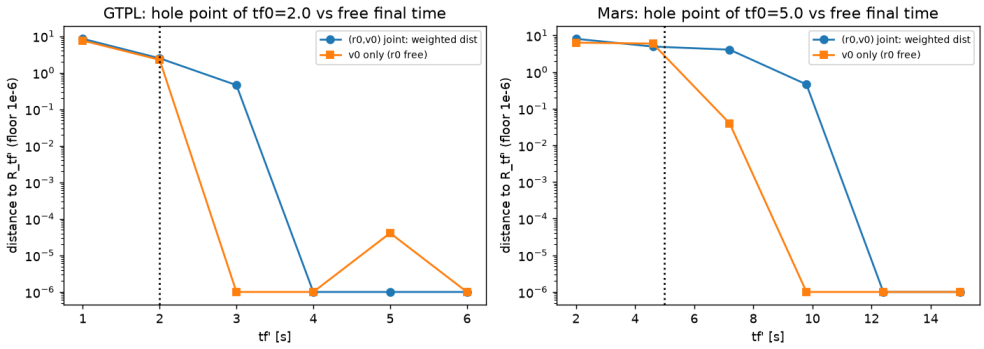


Figure 3: Free final time: the Theorem B hole point at $t_f = 2$ s (an underground datum) is reached at horizons $t'_f \geq 4$ s. The free-final-time obstruction therefore needs the dive-then-brake face and a compact window (Theorem 3.11). Numerical.

Table 1: Size of the nonconvexity at physically admissible data ($r_z \geq 0$), motivating vehicle. All entries numerical (upper bounds on true distances, uncertified global optima) except where marked.

Face / instance	Model	t_f [s]	Gap
B* face, rigorous bound (Theorem 3.7)	M1	7	$\geq 1.3 \times 10^{-5}$ (proved)
B* face, rotation midpoint	M1	4 / 8	≈ 0.010 / 0.007 (weighted m/s)
Slice (S), dive-then-brake	M1	4 / 8	0.016 / 0.009 m in ℓ; offsets 1.5–2.4 m
Pure-dive face (hop, $r_{0z} = 0$)	M1	4 / 8	0.65 / 2.94 (weighted m/s)
Hop landing-set hole	M1	4	up to 4.99 m
Hop landing-set hole	M3 (TVC)	4	1.74 m
Dive-then-brake, upright at rest	PT (6-DOF)	5 / 7	0.028 / 0.033 m
In-air descent (50 m, -5 m/s)	M1	7	none detected (≤ 0.016 m)

5.4 What 3-DOF LCvx asks of the vehicle

The current guidance stack of the motivating vehicle solves a 3-DOF LCvx SOCP and hands only the thrust magnitude to the attitude tracker. Its stored solutions contain thrust-direction jumps of 14.8° within 24 ms and 180° azimuth flips at zero-order-hold switches. These demand roughly 10^3 rad/s^2, against 1–3.5 rad/s^2 available. The relaxation is also not tight on the stored minimum-throttle arc (296 N delivered against $\sigma m = 400$ N), consistent with Theorem 4.2, which places lossiness at minimum throttle. Together with Theorem 4.4 (an upright body has effective cone 0), this shows that the 5-DOF/6-DOF coupling is a first-order issue for this vehicle, independent of the size of the nonconvexity.

6 Machine-checked formalization

The obstruction is short but easy to misstate: data entry, terminal attitude, state constraints and the sign conventions of the dive-then-brake face all matter. We therefore formalized its core in Lean 4 with Mathlib [24, 25] (toolchain `v4.36.0-rc2`; project `lean/LcvxObstruction`, namespace `LcvxObstruction`). The project contains no `sorry` and no `axiom`. For every headline theorem, `#print axioms` reports exactly

```
[propext, Classical.choice, Quot.sound],
```

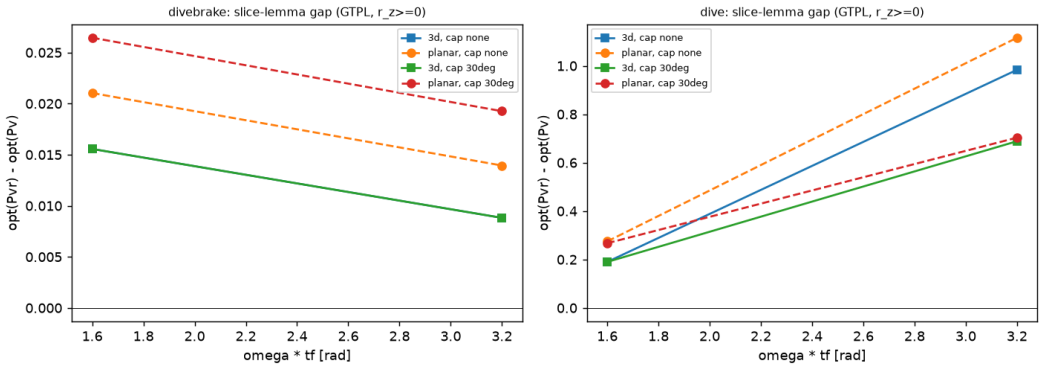


Figure 4: Slice-lemma gaps $\mathrm{opt}(P_{vr}) - \mathrm{opt}(P_v)$ under $r_z \geq 0$ for the dive-then-brake face (left; metres in ℓ) and the pure-dive face (right; m/s), planar versus 3-D (coning), with and without a 30° tilt cap. A strictly positive gap means that the straight-below landing is infeasible while an offset landing is feasible. Numerical.

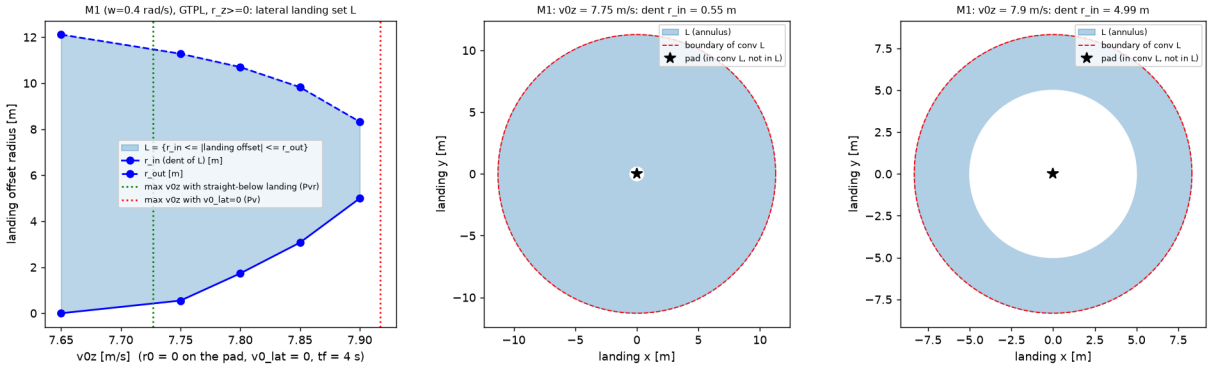


Figure 5: Pad-launched hop ($r_0 = 0$, $v_{0,h} = 0$, $t_f = 4$ s, M1, $r_z \geq 0$): the reachable landing-offset set L is an annulus. The pad lies in $\mathrm{conv}\, L$ but not in L once $v_{0z} \gtrsim 7.73$ m/s. By Theorem D every state-linear relaxation of the minimum-landing-error problem reports error 0 at such data. Numerical; the hole is not certified.

the standard axioms of classical Lean mathematics. This was checked after a full `lake build` at commit `eb2f850`.

6.1 What is formalized

Table 2 lists the formal theorems that correspond to statements in this paper. A row-by-row faithfulness map from each Lean definition to its mathematical meaning is maintained in `lean/MODEL.md`. Its main conventions are listed next.

- $\mathbb{R}^3$ is `EuclideanSpace ℝ (Fin 3)`. Angles are Mathlib's `InnerProductGeometry.angle`. Integrals are Bochner integrals over $[0, t_f]$.
- `UniformConvexRelaxation Θ Y` is a convex set $C \subseteq \Theta \times Y$ for arbitrary real vector spaces Θ, Y. `IsLossless F` is the feasibility form of Theorem 2.4(a), and `IsLosslessFor J F` is the optimization (LCvx) form (b) for an arbitrary cost J, including attainment.

Table 2: Lean theorems corresponding to results in this paper. All compile without `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound` only.

Paper	Lean name (file)	Formal scope
Theorem 3.1, Theorem 2.5	`theoremA`, `not_isLossless_of_not_convex`, `not_isLosslessFor_of_not_convex`, `IsLosslessFor.isLossless` (`Abstract`)	any Θ, Y, any cost
Spherical triangle inequality and rigidity	`inner_triangle`, `inner_triangle_eq_lateral`, `lateral_eq_of_inner_ge` (`Sphere`)	inner-product form with equality case
Theorem 3.4 (variant F)	`not_convex_reachSet`, `not_convex_landingSet`, `no_lossless_relaxation_5dof`, `no_losslessFor_relaxation_5dof`, `landingSet_nonempty` (`Nonconvex`, `Main`)	$\delta + \omega t_f \le \pi/2$, any continuous $a_1 > 0$, any g, no state constraints, $b(t_f)$ free
Theorem 3.7 (qualitative)	`VStar_not_mem_WSet`, `witness_mem_glideSlope`, `not_convex_constrainedLandingSet`, `exists_switchTime`, `no_lossless_relaxation_constrained_5dof`, `no_losslessFor_relaxation_constrained_5dof`, `constrainedLandingSet_nonempty` (`DiveBrake`, `Main`)	constant acceleration bounds ($\alpha = 0$), $g_0 < a_{\max}$, every $t_f > 0$, every $\mathcal{X} \supseteq$ glide-slope set, $b(t_f)$ free
Theorem 3.12 (abstract)	`StateLinearRelaxation.convexHull_subset`, `minError_lossy`, `convex_superset_gap` (`TheoremD`)	abstract; concrete instance not formalized
Theorem 3.10 (abstract)	`not_lineSearch_lossless`, `symmetric_supply`, `not_lineSearch_lossless_of_symmetry` (`FreeTime`)	abstract (i)–(ii); sorry-free (commit `517f1bb`); axiom check after next full build

- `AttitudeAdmissible tf` ω δ `d b` encodes M1 with variant (F): $b(0) = e_z$, unit vectors, $\angle(b(s), b(s')) \le \omega|s - s'|$, measurable unit d, gimbal $\angle(d, b) \le \delta$. Pointwise constraints are imposed for every s, which changes no integral. The admissible thrust accelerations form an abstract class `Acc`. Theorem B assumes only $a \ge a_1$ for a continuous $a_1 > 0$ (so it covers mass depletion through $a_1(s) = \rho_1/(m_0 - \alpha\rho_1 s)$). Theorem B* assumes a box $a_{\min} \le a \le a_{\max}$ with constant bounds, which is exact for constant mass.
- Positions and velocities are given in closed (Cauchy) form, $r(t) = r_0 + tv_0 + \frac{t^2}{2}g + \int_0^t (t - s)\, a\, d$.
- `StateLinearRelaxation` is Theorem 2.6 with an affine readout and a convex penalty that is nonnegative on C.

6.2 The PMP-as-hypothesis convention

Two external theorems appear on the positive side, and each enters only as an explicit hypothesis of the formal statement, never as an axiom. The Pontryagin maximum principle is a structure `PMPConclusion` whose fields are the adjoint equations and the a.e. minimum condition of (H-PMP). Aumann's theorem enters as `AumannInstance`, the single instance used in Theorem 4.1. A reader can therefore check exactly which conclusions depend on which external results. The positive-side files (`Positive/`: pointwise cap lemmas, primer geometry, Theorem 4.2 under `PMPConclusion`, discrete KKT and purification, Theorem 4.4 pointwise and trajectory-level convexity, Theorem 4.1 feasibility under `AumannInstance`) are summarized in `lean/POSITIVE.md`. There the Lean form is weaker than the paper in several places, which that file lists explicitly: the adjoint ODEs are assumed globally differentiable; C-S(2) is constant-mass only; "P_0-optimal" is reduced to pointwise feasibility; C-E is discrete-time; and C-W is constant-mass and unconstrained.

TODO / placeholder (lean, lean-s, lean-positive): formalizations still pending

Not yet committed or not yet axiom-audited at the time of writing:

- `SixDOF.lean` (Theorem 3.19): `not_convex_landingSet6`, `not_convex_constrainedLandingSet6`. Status per [lean]: PENDING.
- `Slice/` (lean-s): `periodic_wirtinger`, `wirtinger_scalar`, `wirtinger_lipschitz` (Theorem 3.13); `master_inequality`, `dent` (master inequality of Theorem 3.14, constant mass); `mixing`, `aOpt`, `no_valueLossless_of_strict` (Theorem 3.15). Lemma S itself (the witness and $0 \notin \operatorname{cl} L$) and Theorem B* beyond the qualitative constant-mass form are not formalized.
- `Positive/` (lean-positive): built locally, but the axiom audit output in `lean/POSITIVE.md` is a placeholder, and the files are not yet in the root import list.

To fill in: move each item into Table 2 once it is committed with a clean `#print axioms`.

6.3 What is not formalized

Mass depletion in Theorem B*; the terminal-attitude variants (T) and (T′); free final time for B* (the abstract line-search lemma is formalized, the supply of its hypotheses from B* is not); all quantitative gaps (Theorem 3.5, the B* bound, ε-losslessness); 6-DOF and torque models; the centre-of-percussion identity; $\rho_1 = 0$; the SDP results; the equivalence of the closed-form trajectory with $r_0 + \int v$ (standard Fubini). The physical scope caveat of Section 3.3 also applies to the formal Theorem B: its witnesses start below ground for a vehicle that cannot hover at minimum throttle. The formal Theorem B* is the statement that covers the altitude-constrained problem.

7 Discussion

7.1 Implications for guidance design

1. *Do not expect a single LCvx-type SOCP for attitude-rate-limited landing.* Within the class of convex programs on the physical state with affine data entry, exactness is impossible, and this cannot be fixed by adding slacks, cuts, lifted variables, discretization or a line search over t_f (Theorems 3.1, 3.7, 3.10 and 3.12). Any such program, if it contains all genuine trajectories, accepts boundary data that no flyable trajectory can land from. The flight software then has no way to detect infeasibility from the convex program alone.

2. *Outer attitude loop, exact convex inner problem.* The nonconvexity lies entirely in the choice of attitude profile (Theorem 4.3). For a fixed attitude profile, the time-varying-cone LCvx is exact (Theorems 4.1 and 4.2). On a pure-TVC vehicle the problem given the attitude is *exactly* convex, with no relaxation (Theorem 4.4). A natural architecture is therefore an outer search over a low-dimensional family of attitude profiles (or a sequential update of b) around an exact convex inner problem. Every inner solution is then genuinely flyable, and the inner problem certifies infeasibility for the given profile.

3. *Certificates.* The degree-2 SDP lift gives a rigorous lower bound on the minimum landing error and certifies infeasibility of exact landing when positive. Its global optimality can be certified a posteriori (rank 3, stress test), and a priori for coplanar targets (Theorems 4.7 and 4.8). It is a useful companion to a local or sequential solver, not a replacement.

4. *How much does it matter?* For in-air descents of a small lander, the exhibited nonconvexity is centimetre-scale, and 3-DOF guidance with an attitude-aware outer loop may lose little in value. The large effects are elsewhere. First, near lift-off (pad-launched hops) the landing set can have a metre-scale hole. Second, 3-DOF LCvx solutions ask for thrust-direction changes far beyond the vehicle's angular acceleration (Section 5). Third, on a pure-TVC vehicle the 3-DOF optimum is essentially never exactly flyable. The "upright body plus pointing cone" 3-DOF model has effective cone zero.

5. *Tilt-angle rate limits.* Constraining only the tilt angle from vertical, with free azimuth, removes rigidity and appears to give convex landing sets numerically. It is not SOC-representable either, and it is not the physical constraint of a rigid body.

7.2 Open problems

1. *Pure TVC, upright landing at rest, with altitude constraint.* This is the actual small-lander case. It is numerically nonconvex (2.8–3.3 cm at $t_f = 5$–7 s; a 1.7 m hole at hop data), with a computer-assisted proof in progress (Section 3.8.1). Mass depletion and gimbal-rate limits remain open even there.

2. *Rigor at realistic constants.* For our vehicle, Lemma S (hard soft landing, per instance) and the hop-instance hole are numerical. The available quadratic, Wirtinger and S-procedure certificates are provably too loose. A two-dimensional rigidity argument or validated branch-and-bound is needed. The $\rho_1 = 0$ landing-set result with altitude constraint is open at $T/W \approx 1.27$.

3. *CSD2′.* Is the degree-2 lift exact in continuous time with fixed thrust, or for the convexified-lens and $\delta = 0$ discretizations? Is there a tractable exact lift with variable throttle (e.g. the separate-axis lift CSD3)?

4. *Minimum-fuel value-losslessness.* Is $V = V^{**}$ on F for minimum fuel? Theorem A-opt reduces this to a convexity question, and numerics show no violation. A positive answer would still not give an LCvx in the detection-sound sense.

5. *Long horizons and coning.* At long horizons with tilt caps, coning fills most but not all of the dents. Whether nonconvexity persists for all horizons in the tent variant, and at faces with long capped plateaus, is open.

6. *Quantitative state-constrained gaps.* The rigorous B* gap ($\sim 10^{-5}$ m) is far below the numerical one ($\sim 10^{-2}$). A sharp quantitative version would settle how much a guidance designer actually loses.

Reproducibility. Proof notes (`math/`), the Lean project (`lean/`), numerical scripts and outputs (`numerics/`) and the claims registry (`comms/CLAIMS.md`) are kept in the project repository. `paper/CLAIMS-MAP.md` maps every theorem of this paper to its proof file, registry row and Lean name.

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A Proofs of Theorems B and Q

Adapted from project note `math/02` §§1–5. Throughout, M1 of Section 2.2.

Lemma A.1 (Spherical triangle inequality, inner-product form)**.** *Let $x, y, z \in S^2$, $\alpha, \beta \in [0, \pi]$, $\alpha+\beta \le \pi$, $\langle x, y\rangle \ge \cos\alpha$, $\langle y, z\rangle \ge \cos\beta$. Then (a) $\langle x, z\rangle \ge \cos(\alpha+\beta)$. (b) If $\langle x, z\rangle = \cos(\alpha+\beta)$ and $\alpha+\beta < \pi$, then $\langle x, y\rangle = \cos\alpha$, $\langle y, z\rangle = \cos\beta$, and, if $\beta \in (0, \pi)$, $x = \cos\alpha\, y - \frac{\sin\alpha}{\sin\beta}(z - \cos\beta\, y)$.* [Lean-checked]

Proof. Write $p = \langle x, y\rangle = \cos a$, $q = \langle y, z\rangle = \cos c$ with $a \le \alpha$, $c \le \beta$, and $x' = x - py$, $z' = z - qy$, both orthogonal to y. Then $\langle x, z\rangle = pq + \langle x', z'\rangle \ge \cos a \cos c - \sin a \sin c = \cos(a + c) \ge \cos(\alpha + \beta)$. Equality forces $a = \alpha$, $c = \beta$ (strict monotonicity of cos on $[0, \pi]$) and $\langle x', z'\rangle = -\|x'\|\|z'\|$, i.e. x' is a nonpositive multiple of z'. □

Corollary A.2 (Extremal thrust direction)**.** *If $\langle d, b\rangle \ge \cos\delta$, $\langle b, e_z\rangle \ge \cos\varphi$, $\varphi \in (0, \pi)$, $\delta + \varphi < \pi$, then $d_z \ge \cos(\delta + \varphi)$. Equality implies $b = \cos\varphi\, e_z + \sin\varphi\, u$ for a horizontal unit u and $d = \cos(\delta + \varphi)e_z + \sin(\delta + \varphi)u$.*

Lemma A.3 (Rigidity)**.** *If $b(s) = \cos\psi\, e_z + \sin\psi\, u$, $b(s') = \cos\psi'\, e_z + \sin\psi'\, u'$ with $\psi, \psi' \in (0, \pi)$, u, u' horizontal unit vectors, and $\langle b(s), b(s')\rangle \ge \cos(\psi - \psi')$, then $u = u'$.* [Lean-checked]

Proof. $\langle b(s), b(s')\rangle - \cos(\psi - \psi') = \sin\psi \sin\psi'(\langle u, u'\rangle - 1) \ge 0$ forces $\langle u, u'\rangle = 1$. □

Proof of Theorem 3.4. The extreme arcs (4) are admissible: $\langle b_u(s), b_u(s')\rangle = \cos(\varphi(s) - \varphi(s'))$ and φ is ω-Lipschitz; the gimbal angle is δ; $a = a_1$ at constant minimum throttle. So $P_{\pm u} \in R$ and $M \in \operatorname{conv} R$. Suppose $M = \int a\, d$ for an admissible (T, d, b). In both variants $\omega t_f \le \pi$ and $0 \le \delta + \varphi \le \pi/2$.

Step 1. (AL_ω) with $s' = 0$ (and $s' = t_f$ in variant T) gives $b_z(s) \ge \cos\varphi(s)$. By Theorem A.1(a), $d_z \ge \cos(\delta + \varphi) \ge 0$, and with $a \ge a_1$, $a\, d_z \ge a_1 d_z \ge a_1 \cos(\delta + \varphi)$ a.e. *Step 2.* Integrating, $M_z \ge \int a_1 \cos(\delta + \varphi) = M_z$, so the nonnegative integrand $a\, d_z - a_1 \cos(\delta + \varphi)$ vanishes a.e. Since $a_1 > 0$, $d_z = \cos(\delta + \varphi)$ a.e. *Step 3.* By Theorem A.2, for a.e. $s \in (0, t_f)$, $b_z(s) = \cos\varphi(s)$ and $d(s) = \cos(\delta + \varphi)e_z + \sin(\delta + \varphi)u(s)$, where $u(s)$ is the azimuth of $b(s)$. *Step 4.* $b_z - \cos\varphi$ is continuous and vanishes a.e., hence everywhere, so $u(\cdot)$ is defined and continuous on $(0, t_f)$. *Step 5.* In variant F, for $0 < s < s'$, (AL_ω) gives $\langle b(s), b(s')\rangle \ge \cos(\varphi(s') - \varphi(s))$, and Theorem A.3 gives $u(s) = u(s')$. In variant T apply this on $(0, t_f/2]$ and on $[t_f/2, t_f)$ and chain through $t_f/2$. So $u \equiv u_0$. *Step 6.* $I_h = (\int a \sin(\delta + \varphi))u_0$ with coefficient $\ge \int a_1 \sin(\delta + \varphi) > 0$, contradicting $M_h = 0$. Nonconvexity of F_{t_f} follows from Theorem 3.3, and impossibility from Theorem 3.1. □

Lemma A.4 (Support bound)**.** *For any admissible control (no boundary condition on b needed) and τ with $\delta + \omega \max(\tau, t_f - \tau) \le \pi/2$, $\|I\| \ge \langle I, b(\tau)\rangle \ge G(\tau) = \int_0^{t_f} a_1(s)\cos(\delta + \omega|s-\tau|)\,ds$.*

Proof. (AL_ω) gives $\langle b(s), b(\tau)\rangle \ge \cos(\omega|s-\tau|)$, and Theorem A.1(a) gives $\langle d(s), b(\tau)\rangle \ge \cos(\delta + \omega|s-\tau|) \ge 0$. Multiply by $a \ge a_1$ and integrate. □

Proof of Theorem 3.5. For $I \in R$, $\|I - M\| \ge \|I\| - \|M\| \ge G(t_f/2) - M_z$ by Theorem A.4. With $c(x) = \cos(\delta + \omega x)$, strictly decreasing on $[0, t_f]$, put $h(s) = c(|s - t_f/2|) - c(s)$. Then $h \ge 0$ for $s \ge t_f/4$ and $h < 0$ for $s < t_f/4$. Since a_1 is nondecreasing, $(a_1(s) - a_1(t_f/4))h(s) \ge 0$, so $G(t_f/2) - M_z = \int a_1 h \ge a_1(t_f/4)\int h$. A direct computation, using $\sin x + \sin(x+2k) = 2\sin(x+k)\cos k$, gives $\int_0^{t_f} h = \frac{4}{\omega}\sin(\delta + \frac{\omega t_f}{2})\sin^2\frac{\omega t_f}{4}$. The sign condition holds if $\delta + \omega t_f \le \pi/2$ (nonnegative integrand), or if $\alpha = 0$, where $M_z = \frac{a_1}{\omega}(\sin(\delta + \omega t_f) - \sin\delta) \ge 0$. The relaxation statement is Theorem 3.2(ii) together with Theorem 3.3. □

B Proof of Theorem B*

Adapted from project note `math/02` §10, which survived adversarial review (`math/01`). Notation: $a_{10} = \rho_1/m_0$, $a_{\max} = \rho_2/(m_0 - \alpha\rho_2 t_f)$, $A_b = \rho_2/m_0 - g_0$, $w(s) = s - s^*$. For a control c put $W(c) = \int w\,a\,d$, $X(c) = \int_0^{s^*}(s^* - s)a\,d$, $B(c) = \int_{s^*}^{t_f}(s - s^*)a\,d$, so $W = -X + B$. By (5), $\Lambda(\theta) = W(c) + \kappa e_z$ with $\kappa = g_0 t_f(s^* - t_f/2)$, for any control c landing from θ.

Proof of Theorem 3.7 (qualitative part). Step 1 (direction bound for a fixed throttle). Fix T, hence $a = a_T > 0$. On $[0, s^*]$, $b_z \ge \cos\omega s$ and $\delta + \omega s \le 2\delta < \pi$, so $d_z \ge \cos(\delta + \omega s)$. On $(s^*, t_f]$, $d_z \le 1$. Since $w < 0$ before s^* and $w > 0$ after,

$$W_z(c) \le V(T) := -\int_0^{s^*}(s^* - s)a_T\cos(\delta + \omega s)\,ds + \int_{s^*}^{t_f}(s - s^*)a_T\,ds.$$

Step 2 (attained for every T). The dive-then-brake control c_u^T uses $b = \cos(\omega s)e_z + \sin(\omega s)u$ and $d = \cos(\delta + \omega s)e_z + \sin(\delta + \omega s)u$ on $[0, s^*]$, then $b \equiv b(s^*)$ and $d = e_z$. It is admissible, because by (S1) $\angle(e_z, b(s^*)) = \omega s^* \le \delta$. In variant (T′), return b meridionally to e_z on $[s^*, 2s^*]$. The gimbal angle stays $\le \delta$. *Step 3 (late throttle).* Replacing T by ρ_2 on $(s^*, t_f]$ leaves a unchanged before s^* and does not decrease it after. So $V(T) \le V(T^\sharp)$, strictly unless $T = \rho_2$ a.e. late. *Step 4 (outer maximum).* $V^* = \sup_T V(T)$ is attained, explicitly for $\alpha = 0$ ($T^* = \rho_1$ then ρ_2), and for $\alpha > 0$ by weak-$*$ compactness of the throttle set and weak-$*$ continuity of V. *Step 5 (state constraints).* On the brake $d = e_z$, so $v_h = r_h = 0$ there and $\dot v_z \ge A_b$. Hence $v_z(t) \le -A_b(t_f - t)$ and $r_z(t) \ge \frac{1}{2}A_b(t_f - t)^2$. On the dive $\dot v_z \ge -g_0$, so $v_z < 0$ by (S2). Thus r_z is decreasing, hence positive before t_f. Laterally $\|r_h\| \le \frac{1}{2}a_{\max}s^{*2}$, and (S3) gives the glide slope. By rotation invariance, $R_\pi\theta^*$ is landed by $c_{-u}^{T^*}$. *Step 6 (midpoint infeasible).* $\Lambda(\theta_m) = (V^* + \kappa)e_z$. If $\theta_m \in F_{t_f}$ via c, then $V^* = W_z(c) \le V(T) \le V^*$, so all inequalities are equalities a.e. On the dive, $d_z = \cos(\delta + \omega s)$, so Theorems A.2 and A.3 give a constant azimuth u_0. On the brake, $d = e_z$. Then $W_h(c) = -(\int_0^{s^*}(s^* - s)a_T\sin(\delta + \omega s))u_0 \ne 0$, contradicting $W_h(c) = \Lambda_h(\theta_m) = 0$. □

Proof of the quantitative bound ($\delta \le \pi/4$). Let c land from $\theta \in F_{t_f}$ and put $\eta = V^* - W_z(c) \ge \eta_{dive} + \eta_{brake}$, the deficits relative to Step 1. Let $\tau = \vartheta s^*$ with $\vartheta = a_{10}\sin\delta/(4a_{\max}) \le \frac{1}{4}$, and $c_\tau = b(\tau)$, of tilt $\beta \le \omega\tau$. (i) As in Theorem A.4, $\langle X, c_\tau\rangle \ge G(T) := \int_0^{s^*}(s^* - s)a_T\cos(\delta + \omega|s - \tau|)$, since $\delta + \omega s^* \le \pi/2$. (ii) With $c_\tau = \cos\beta\,e_z + \sin\beta\,n$: $\sin\beta\|X_h\| \ge G - X_z \ge G - Z_0 - \eta_{dive}$, where $Z_0 = \int_0^{s^*}(s^* - s)a_T\cos(\delta + \omega s)$. (iii) The mean value theorem gives $G - Z_0 \ge \omega\tau\gamma_X$, using $(1 - \vartheta)^2 - \frac{1}{4} \ge \frac{5}{16}$. (iv) Since $\sin\beta \le \omega\tau$: $\|X_h\| \ge \gamma_X - c_1\eta_{dive}$ with $c_1 = 1/(\omega\tau)$. (v)

$\|d_h\| \le \sqrt{2(1-d_z)}$ and Cauchy–Schwarz give $\|B_h\| \le \sqrt{2W_b\eta}$. (vi) $\|\Lambda(\theta)-\Lambda(\theta_m)\|^2 = \|W_h\|^2+\eta^2$ and $\|W_h\| \ge \gamma_X - c_1\eta - \sqrt{2W_b\eta}$, so $\|\Lambda(\theta-\theta_m)\| \ge \eta_0$. Finally, $\|\Lambda\| = \sqrt{1+s^{*2}}$. □

Proof of the free-final-time window variant. $\ell^*(t) = \sup_{F_t} \ell = V_t^* + \kappa(t)$ is continuous in t, so it attains its maximum on $[t_a, t_b]$ at some t°. If the union were convex, $\theta_m(t^\circ) \in F_{t'}$ for some t' with $\ell(\theta_m) \ge \ell^*(t')$. Then θ_m would be a maximizer at horizon t', and Step 6 at that horizon forces $\Lambda_h \ne 0$. Note that Λ does not depend on the horizon. □

C Proof of Theorem S (Lemma S in a regime)

Adapted from project note `math/07`, which survived adversarial review (`math/01` §9). Model M1, $b(t_f)$ free, $T := t_f$, $\Omega = \omega T$, $\tau(s) = \delta + \omega s$, $\bar\tau = \delta + \Omega$, $a_{10} = \rho_1/m_0$, $a_2 = \rho_2/(m_0 - \alpha\rho_2 T)$, $\varepsilon_m = a_1(T)/a_{10} - 1$. The lateral force is $F = a\, d_h$, and the *gain* of a control is $\int a_1 - I_z$, the vertical impulse wasted relative to upright minimum throttle. P is the $L^2((0,T);\mathbb{R}^2)$ projection onto affine functions. It depends only on the moments $m(F) = (\int F, \int sF)$, and π_m denotes the affine function with moments m. In particular $\|\pi_{(0,q)}\|^2 = 12|q|^2/T^3$.

Landing from vertical data. From $\zeta = (he_z, v_{0z}e_z)$ a trajectory lands at $(p,0)$ with $v(T) = 0$ iff $I_h = 0$, $I_z = g_0T - v_{0z}$, $\int sF = -p$ and J_z is fixed by ζ. Hence *every trajectory from ζ has the same gain*, and $m(F) = (0, -p)$.

Proof of Theorem 3.13. Let $\tilde f = f - f(0) - \frac{s}{T}(f(T) - f(0))$. Then $f - Pf = \tilde f - P\tilde f$, and $\|\tilde f - P\tilde f\| \le \|\tilde f - \bar{\tilde f}\|$ for the mean $\bar{\tilde f}$. Since $\tilde f(0) = \tilde f(T) = 0$, $\tilde f$ extends periodically, and the periodic Wirtinger inequality (Parseval) gives $\|\tilde f - \bar{\tilde f}\|^2 \le (T/2\pi)^2\|\dot{\tilde f}\|^2 = (T/2\pi)^2(\|\dot f\|^2 - |f(T) - f(0)|^2/T)$. All steps are componentwise. □

Master inequality. With $c_0 = 1/(1+\cos\delta)$, $\delta_e = 2\sin(\delta/2) + \varepsilon_m \sin\bar\tau$, $E_4 = a_{10}(1+\varepsilon_m)^2 \int_0^T (\frac{1}{1+\cos\tau(s)} - c_0)\sin^2\tau(s)\, ds$, $B = a_{10}c_0T(\frac{\Omega}{2\pi} + \delta_e)^2 + E_4$, $A' = c_0/a_{10}$, $\kappa_0 = \cos\bar\tau/((a_2 - a_{10})\sin^2\bar\tau)$ and $C_m = A'\kappa_0/(\kappa_0 - A')$ ($C_m = A'$ if $\rho_1 = \rho_2$):

Proposition C.1 (Master inequality [`math/07, Prop. M`])**.** *If $\bar\tau < \pi/2$ and $\kappa_0 > A'$, every admissible M1 trajectory (any 3-D attitude path, throttle, $\alpha \ge 0$) satisfies* gain $\le B + C_m\|\pi_{m(F)}\|^2$. *[proved; survived attack]*

Proof sketch. (1) A pointwise majorant $-a\, d_z \le -a_1 + a_1\bar c|d_h|^2 - (a - a_1)\cos\tau$ with $\bar c = 1/(1+\cos\tau)$ splits off the throttle surplus $K = (a - a_1)d_h$, which costs at least $\kappa_0\|K\|^2$. (2) Normalize the mass. (3) Eliminate K through its projection ($PK = \pi_m - a_{10}Pv$) by maximizing a concave quadratic. (4) The only nonconvex step bounds the non-affine part of the lateral tilt $\beta = b_h$, which is ω-Lipschitz, by Theorem 3.13: $\|P^\perp\beta\| \le \omega T^{3/2}/(2\pi)$. The gimbal and mass corrections are bounded pointwise by δ_e. □

Witness. A planar single-crossing S-turn at minimum throttle: tilt $\varphi(s) = \omega s$ for $s \le t_1$ and $\omega(2t_1 - s)$ after, the thrust deflected outward by δ ($\chi = \varphi + \delta\,\mathrm{sgn}\,\varphi$), and t_1 chosen by the intermediate value theorem so that $\int a_1 \sin\chi = 0$ (so $I_h = 0$). Its mean-square lateral tilt is $(3\sqrt{2} - 4)/6 \cdot \Omega^2 \approx 0.0404\,\Omega^2$ in the small-angle limit ($t_1 = (1 - 1/\sqrt{2})T$). This exceeds the Wirtinger cap $\Omega^2/(4\pi^2) \approx 0.0253\,\Omega^2$. The witness does not return to zero tilt, which is why it beats every path whose two lateral moments vanish.

Proof of Theorem 3.14. (1) Along the witness $\dot{v}_z = a_1 \cos\chi - g_0 > 0$ by (S1), so $v_z < 0$ on $[0, T)$ and $r_z > 0$ before T. Rotations about e_z preserve ζ_w, so $L(\zeta_w)$ contains the circle $|p| = |p_w|$. (2) Every trajectory from ζ_w landing at $(p, 0)$ has gain gain_w and $m(F) = (0, -p)$, so Theorem C.1 gives $\text{gain}_w \le B + 12C_m|p|^2/T^3$, i.e. $|p| \ge r_S$. (3) Apply Theorem 3.12 with $\varphi(x) = \|x_r\| + \iota_{\{x_v=0\}}(x)$; since $0 \in \operatorname{conv} L(\zeta_w)$, the relaxed value is 0. □

Table 3: Certified instances of Theorem 3.14 for a demonstration vehicle ($m_0 = 100$ kg, $\rho_1 = 1200$ N, $\rho_2 = 2500$ N, $\alpha = 0$), from `math/07` §4.

δ	T [s]	Ω [rad]	margin M [m/s]	dent r_S [m]	witness offset $\lvert p_w\rvert$ [m]
0	2	0.40	1.39×10^{-2}	0.45	0.94
0	4	0.40	2.78×10^{-2}	1.80	3.74
5°	2	0.30	1.57×10^{-2}	0.48	1.79
10°	2	0.20	1.06×10^{-2}	0.39	2.59
10°	4	0.20	2.11×10^{-2}	1.57	10.38

With $\alpha = 5.66 \times 10^{-4}$ the $\delta = 5°$ row fails narrowly, because ε_m enters δ_e linearly. That is a weakness of the constants, not of the geometry. For $\delta \ge 15°$ the analytic constants do not certify; a discretized S-procedure SDP suggests $\delta = 15°$ holds with an 8% margin (numerical).

D Further proof sketches

D.1 Theorem P0-F ($\rho_1 = 0$)

Adapted from `math/possibility-rho0` §2. Set $\rho_1 = 0$, $\alpha = 0$, $a_2 = \rho_2/m_0$, units $\omega = a_2 = 1$, $T = \omega t_f$. For fixed b the pointwise control set $U(b) = \{u : ormu \le a_2,\ \langle u, b\rangle \ge ormu \cos\delta\}$ (convex, containing 0) has support function $a_2\|p\|\, c(\angle(p, b))$ with $c(x) = 1$ for $x \le \delta$, $\cos(x - \delta)$ for $\delta \le x \le \delta + \pi/2$, and 0 beyond. K is compact and star-shaped about 0, so it is convex iff every exposed face lies in K.

Take $\hat{p} = (\cos\delta, 0, -\sin\delta)$ and $\sigma \in (0, t_f)$, primer $p(s) = (s - \sigma)\hat{p}$, face functional $\ell_\sigma = \langle \hat{p}, \int (s - \sigma) a\, d\rangle = \langle \hat{p}, r_0 + \sigma v_0\rangle + \text{const}$. With $\beta = \phi_0 - \angle(\hat{p}, b)$ (an ω-Lipschitz tilt toward $\hat{p}$, $\beta(0) = 0$) one has $\ell_\sigma \le a_2 P[\beta]$ for an explicit integral P, with equality along great circles through e_z and $\hat{p}$. For fixed $\beta^* = \beta(\sigma)$ the optimal profile is the lower envelope before σ and the upper envelope after, so $h_K = a_2 \max_{\beta^*} V(\beta^*)$ with $V = E + L_t$. Here E is nonincreasing (strictly on $[-T, 0]$, because $T < \pi/2 - 2\delta$), and L_t vanishes iff $\beta^* \le -\tau$, $\tau = T - \sigma$. Let $A(\sigma) = E(-\sigma)$ and $B(\sigma) = \max_{[-\tau,\sigma]} V$. At $\sigma = T$, $B < A$. At $\sigma = T/2 = h$, $A - V(h) \le -f(h)$ with $f(h) = \sin 2h - h\cos 2h + \sin h - 2h \ge h^3(\frac{1}{2} - \frac{2}{3}h^2) > 0$, so $B > A$. Hence there is a tie $\sigma_c \in (T/2, T)$. The two groups of maximizers are compact, disjoint, and separated by $\langle \hat{p}, I\rangle$: the left group coasts late, while the right group thrusts late and less early. A convex face cannot be a disjoint union of two nonempty compact sets, because convex sets are connected. □

D.2 Theorem C-W (fixed attitude)

Adapted from `math/04` §3. The extreme points of the cap $K_\Gamma(b)$ are exactly its spherical part $S_\Gamma(b) = \{\|T\| = \Gamma,\ \angle(T, b) \le \delta\}$, and $K_\Gamma = \operatorname{conv} S_\Gamma$ (Krein–Milman; Lean: `extremePoints_cap`, `cap_subset_closure_convexHull_sCap`). On each subinterval between prescribed nodes, the state increment is $\int \Psi_k(s) T(s)/m(s)\, ds$ with Ψ_k of full column rank and m fixed by Γ. Aumann's

theorem applied to the multifunction $\Psi_k S_\Gamma(b)/m$ yields a measurable lossless selection with the same increment. Concatenation over subintervals preserves the state at every node. $\square$

D.3 Theorem C-6 (pure TVC given attitude)

With zero roll rate the angular momentum is $J_\perp b \times \dot{b}$, and Euler's equation reads $J_\perp b \times \ddot{b} = L\,T \times b$. Splitting T and $\ddot{b}$ into components along and orthogonal to b gives $\Pi_b T = -k\Pi_b \ddot{b}$. So $T = T_a b + F$ with F determined by b. The gimbal condition becomes $\varkappa \le T_a \tan\delta$, and the throttle condition becomes $\rho_1^2 \le T_a^2 + \varkappa^2 \le \rho_2^2$. Both are intervals in T_a. The dynamics are affine in T_a for fixed b, and the fuel integrand $\|(T_a, \varkappa)\|$ is convex. $\square$